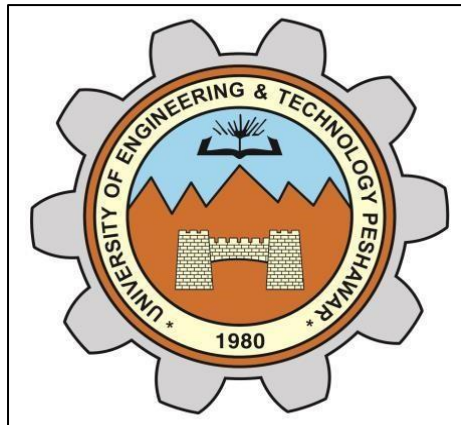


**DESIGN AND OPTIMIZATION OF 4-PORT MIMO
ANTENNA FOR 5G APPLICATIONS AND PREDICTION
OF DIRECTIVITY WITH MACHINE LEARNING
MODELS**



Supervisor

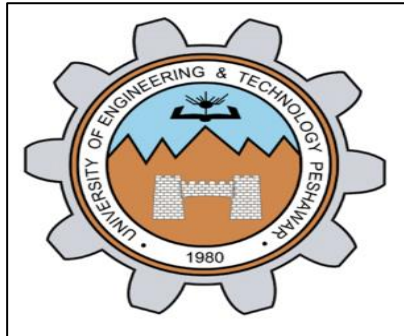
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SESSION 2021-2025**

APPROVAL SHEET



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Session 2021-2025**

ABSTRACT

This research presents a comprehensive design and optimization study of a compact 4-Port MIMO antenna intended for 5G wireless communication applications. The proposed antenna demonstrates outstanding performance characteristics, including a remarkably low return loss reaching up to -58 dB, a high gain peaking at 9.5 dB, and a directivity value as high as 10.1 dB, ensuring efficient impedance matching and superior radiation properties essential for next-generation wireless networks that demand high data throughput and minimal interference. The design process involved careful tuning and iterative optimization to achieve the desired band performance while maintaining high isolation between antenna ports, thereby reducing mutual coupling effects that can degrade MIMO system efficiency. In addition to the antenna engineering aspects, this study incorporates advanced machine learning techniques to predict the antenna's directivity based on its design parameters, significantly reducing the reliance on computationally intensive full-wave electromagnetic simulations traditionally used in antenna design. Various machine learning algorithms were explored, with the Extreme Gradient Boosting (XGBoost) model emerging as the most effective, delivering an impressive prediction accuracy with an efficiency of 99.97%. The success of this model demonstrates the potential for integrating machine learning into antenna design workflows, enabling rapid evaluation and optimization of antenna performance metrics. This hybrid methodology not only accelerates the development cycle but also provides a reliable tool for engineers to explore design variations and enhance 5G MIMO antenna systems efficiently. Overall, this thesis contributes to both practical antenna design for 5G applications and the growing field of machine learning-assisted electromagnetic device optimization.

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List Of Contents

APPROVAL SHEET	2
ACKNOWLEDGMENTS	4
List Of Figures.....	08
List Of Acronyms.....	09
Chapter 01 Introduction	10
1.1 Background and Motivation	10
1.2 Overview of 5G Communication Systems	11
1.3 Importance of Antenna Design in 5G Wireless Systems.....	12
1.3.1 Enhancing Signal Strength and Coverage:	13
1.3.2 Enabling MIMO and Beamforming:.....	13
1.3.3 Supporting Compact and Integrated Designs:	13
1.3.4 Improving Spectral and Energy Efficiency:	13
1.3.5 Adapting to Environmental Challenges:	13
1.4 Basic Microstrip Patch Antenna	14
1.4.1 Structure and Operation of Microstrip Patch Antennas.....	14
1.5 Multiple-Input Multiple-Output (MIMO) Antennas.....	16
1.5.1 Fundamentals of MIMO Technology	16
1.5.2 Advantages of MIMO in Wireless Systems.....	17
1.5.3 Challenges in MIMO Antenna Design	18
1.6 Key Antenna Parameters and Performance Metrics	18
1.6.1 S-Parameters and Return Loss	18
1.6.2 Radiation Pattern.....	19
1.6.3 Directivity	20
1.6.4 Gain.....	21
1.6.5 Radiation Efficiency	22
1.6.6 Voltage Standing Wave Ratio (VSWR)	22
1.6.7 Input Impedance and Impedance Matching.....	23
1.6.8 Polarization	23
1.7 MIMO Antenna Parameters.....	24
1.7.1 S-Parameters (Reflection and Transmission Coefficients).....	24
1.7.2 Isolation and Mutual Coupling	24
1.7.3 Envelope Correlation Coefficient (ECC).....	25
1.7.4 Diversity Gain (DG)	25
1.7.5 Channel Capacity Loss (CCL)	25
1.7.6 Gain and Directivity.....	26
1.8 Machine Learning in Antenna Design and Optimization	26
1.8.1 Introduction to ML Techniques in Antenna Design	26
1.8.2 Benefits of ML in Predicting Antenna Performance	26
1.8.3 Overview of Regression Models Relevant to Antenna Parameter Prediction	27
1.9 Problem Statement and Research Gaps	27
1.9.1 Limitations of Existing MIMO Antenna Designs.....	27
1.9.2 Challenges in Directivity Prediction and Gain Enhancement.....	27
1.9.3 Gaps in Integrating ML Techniques for Antenna Optimization.....	28
1.10 Aims and Objectives	28
1.10.1 Design and Fabrication of Compact 4-Port MIMO Antenna	28
1.10.2 Performance Evaluation: Gain, Directivity, Isolation, Efficiency ...	28
1.10.3 Application of Machine Learning for Directivity Prediction	28
1.10.4 Improvement of Antenna Parameters Using ML-Assisted Methods	28

1.11 Scope of the Thesis	29
1.12 Thesis Organization	29
1.13 Conclusion	29
Chapter 02 Literature Review	31
2.1 Introduction.....	31
2.2 Review of mmWave Antennas for 5G.....	31
2.3 Design Techniques for Compact and High-Gain Antennas.....	33
2.4 State-of-the-Art MIMO Antennas for 28 GHz Band.....	33
2.5 Optimization Methods in Antenna Design	34
2.5.1 Traditional Parametric Sweeps and Simulation.....	35
2.5.2 Introduction to Machine Learning Techniques.....	35
2.5.3 Supervised Learning and Regression Models.....	35
2.5.4 Challenges in Data Collection and Model Training	35
2.6 Application of Machine Learning in Antenna Design and Prediction.....	36
2.7 Research Gaps and Motivation for Current Study	37
2.7.1 MIMO Design Difficulty	37
2.7.2 Complexity of Design and Computational Cost	37
2.7.3 Dataset Compiling for Machine Learning Models	38
2.8 Summary	38
Chapter 03 Methodolgy And Design.....	39
3.1 Design Specifications and Requirements for MIMO Antenna	39
3.2 Substrate Material Selection and Its Impact on Performance.....	40
3.3 Theoretical Background and Design Equations for Microstrip Patch Antennas	42
3.4 Single Element Antenna Design	42
3.4.1 Design Diemensions	43
3.4.1 Patch Geometry and Feeding Mechanism	45
3.4.2 Slot Introduction for Bandwidth Enhancement	45
3.4.3 Feedline Modifications and Optimization	46
3.5 Parametric Studies for Single Element Antenna.....	46
3.5.2 Effect of Substrate Width.....	48
3.5.3 Effect of Patch Length	48
3.6 Transition to Array Antenna	49
3.6.1 2-Element Array Design and Optimization	50
3.6.2 Array Spacing and Its Effect on Gain and Bandwidth.....	50
3.7 Design and Optimization of 4-Port MIMO Antenna	52
3.7.1 Element Arrangement and Spacing	52
3.7.2 Techniques for Mutual Coupling Reduction.....	53
3.7.3 Final Optimized Layout and Dimensions	53
3.8 Simulation Environment and Tools	55
3.9 Machine Learning Approach for Directivity and Gain Prediction	55
3.9.1 Dataset Generation from Parametric Sweeps	55
3.9.2 Input Features and Output Parameters.....	56
3.9.3 Description of Regression Models.....	56
3.9.4 Training, Validation and Performance Metrics	57
3.10 Summary	58
Chapter 4: Results and Discussions.....	59
Introduction.....	59
4.1 Single Element Antenna	59
4.2 Single Element Results	60

4.2.1 Reflection Coefficient (Return Loss) and Resonant Frequency	60
4.2.2 Bandwidth	60
4.2.3 VSWR	61
4.2.4 Efficiency	61
4.2.5 Gain and Directivity	63
4.3 Transition from single element to 2- Element array	65
4.3.1 Design Dimensions and optimization	65
4.3.2 Reflection co-efficient and resonant frequency	67
4.3.3 Bandwidth	67
4.3.4 VSWR	68
4.3.5 Efficiency	69
4.3.6 Gain and Directivity	70
4.4 Design and Analysis of 4-Port MIMO Antenna	73
4.4.1 Motivation and Design Objectives.....	73
4.4.2 Design Dimensions	73
4.4.5 Transmission Co-efficients and Isolation	77
4.4.6 VSWR of MIMO	77
4.4.7 Gain And Directivity of MIMO	79
4.4.8 Envelope Correlation Coefficient (ECC).....	81
4.4.9 Diversity Gain (DG)	82
Calculation of Diversity Gain	82
4.4.10 Dataset Collection	83
4.5 Machine Learning Model Training.....	84
4.5.1 Model Training and Evaluation for Predictive Modeling.....	84
4.5.2 Data Loading and Initial Assessment	85
4.5.4 Feature Selection.....	86
4.5.5 Model Selection	86
4.5.6 Data Splitting	86
4.5.7 Model Training	86
4.5.9 Model Saving and Loading	87
4.5.10 Model Prediction (Inference).....	87
4.5.11 Summary of Machine learning.....	88
4.6 Conclusion	88
Chapter 5	90
Conclusions and Future Work	90
References.....	93

List Of Figures

Figure 1.1 return loss (S11) vs Frequency graph of 5G antenna.....	19
Figure 1.2 3D Radiation pattern of antenna.....	20
Figure 1.3 Directivity & Gain of antenna.....	21
Figure 1.4 Showing different types of polarization.....	23
Figure 3.1 Single Element With dimensions.....	43
Figure 3.2 Effect of feed length on resonance frequency.....	47
Figure 3.3 Effect of substrate width on resonance frequency.....	48
Figure 3.4 Effect of patch length on resonance frequency.....	49
Figure 3.5 Illustrating 2-Element Array.....	51
Figure 3.6 Showing 4-Port MIMO Antenna with mentioned dimension..	54
Figure 4.1 single element design.....	59
Figure 4.2 S11 and frequency Graph.....	60
Figure 4.3 VSWR graph of single element antenna.....	61
Figure 4.4 Power components vs. Frequency	62
Figure 4.5 Directivity polar plot.....	63
Figure 4.6 Gain polar plot.....	64
Figure 4.7 3D View of single element illustrating gain.....	64
Figure 4.8 Directivity and 3D radiation pattern of single element.....	65
Figure 4.9 2-Element Array.....	66
Figure 4.10 2-Element array with dimensions.....	66
Figure 4.11 S11-Frequency graph of 2-Element Array.....	67
Figure 4.12 Bandwidth points marked on S11-freq graph.....	68
Figure 4.13 VSWR of 2-Element array.....	69
Figure 4.14 Showing curves of different powers and losses	70
Figure 4.15 Directivity of 2 element shown in polar plot.....	71
Figure 4.16 Gain of 2 element shown in polar plot.....	71
Figure 4.17 3D radiation pattern of 2-element antenna with gain values	72
Figure 4.18 3D radiation patt.....	72
Figure 4.19 2D view of 4-port MIMO antenna.....	74
Figure 4.20 3D MIMO view with dimensions.....	74
Figure 4.21 2D MIMO view with dimensions.....	75
Figure 4.22 S11 and frequency of MIMO.....	75
Figure 4.23 Illustrating bandwidth points marked in s11 grap	76
Figure 4.24 Showing graph of transmission co-efficients.....	77
Figure 4.25 VSWR of MIMO.....	78
Figure 4.26 Impedance graph.....	78
Figure 4.27 polar plot for directivity.....	79
Figure 4.28 Polar Plot of Gain.....	80
Figure 4.29 3D view of Gain.....	80
Figure 4.30 3D view of directivity.....	81
Figure 4.31 Graph of ECC.....	82
Table 4.1 Comparison of Single Element ,2-Element and MIMO.....	83
Table 4.2 Dataset used for ML Model Training.....	84

Table of Acronyms

Acronym	Definition
ANNS	Artificial Neural Network
CCL	Channel Capacity Loss
CNN	Convolutional Neural Network
DG	Diversity Gain
ECC	Designing antennas that produce low envelope correlation coefficient
EE	Electrical Engineering
FIT	Finite Integration Technique
GPS	Global Positioning Systems
ICSET	IEEE International Conference on System Engineering and Technology
IQR	Interquartile Range
MAE	Mean Absolute Error
MARS	Multivariate Adaptive Regression Splines
MIMO	Multiple Input Multiple Output
ML	Machine Learning
MSE	Mean Squared Error
PCB	Printed circuit board
RMSE	Root Mean Squared Error
TARC	Total Active Reflection Coefficient
VSWR	Voltage Standing Wave Ratio
WLAN	Wireless Local Area Networks
XGB	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Background and Motivation

In recent years, the rapid advancement of wireless communication technology has transformed the way people connect, communicate, and access information. Among these advancements, the introduction and ongoing deployment of fifth-generation (5G) wireless networks represent a significant leap forward in communication capabilities. 5G technology promises ultra-high data rates, extremely low latency, enhanced reliability, and the ability to connect a vast number of devices simultaneously. These features enable a broad range of new applications, including autonomous vehicles, smart cities, virtual and augmented reality, and massive Internet of Things (IoT) deployments.

Antenna design lies at the heart of these advances, as antennas are responsible for the transmission and reception of electromagnetic signals. The performance of a wireless communication system heavily depends on the characteristics of its antennas, such as gain, bandwidth, directivity, efficiency, and isolation in multi-antenna systems. Designing antennas for 5G, especially those operating in millimeter-wave (mmWave) frequency bands (above 24 GHz), is challenging due to increased path losses, signal blockages, and more complex propagation behaviors at higher frequencies.

In particular, Multiple-Input Multiple-Output (MIMO) antenna systems are a key enabling technology for 5G, providing higher data throughput and improved link reliability by exploiting spatial diversity and multiplexing multiple data streams. However, as the number of antenna elements increases within a limited space, issues such as mutual coupling and signal interference arise, which can degrade performance.

To address these challenges, researchers have been exploring innovative antenna structures, materials, and design methodologies. One promising approach is the use of metasurfaces—engineered surfaces with sub-wavelength structures designed to manipulate electromagnetic waves in ways unattainable with traditional materials. Metasurfaces can enhance antenna gain, bandwidth, and isolation, enabling compact

and efficient designs suitable for the tight integration requirements of 5G devices.

Alongside hardware innovations, machine learning techniques have recently gained attention for their ability to model and predict complex antenna behaviors rapidly. Traditional electromagnetic simulations are computationally expensive and time-consuming, especially when exploring large design spaces with many parameters. Machine learning models, trained on simulation or measurement data, can predict antenna performance metrics such as directivity and gain with high accuracy, drastically reducing design iteration time and enabling faster optimization.

This thesis is motivated by the need to develop a compact, high-performance 4-port MIMO antenna for 5G mmWave applications and to demonstrate the effectiveness of machine learning models for predicting antenna directivity. By combining advanced antenna design with data-driven prediction, this research aims to contribute to more efficient and intelligent antenna development workflows, meeting the stringent demands of next-generation wireless networks.

1.2 Overview of 5G Communication Systems

Fifth-generation (5G) wireless communication represents the latest evolution in mobile network technology, designed to meet the growing demand for high-speed, low-latency, and massive connectivity. Unlike its predecessors, 5G is not just an incremental improvement but a transformative platform enabling new use cases and services.

One of the defining features of 5G is its utilization of higher frequency bands, particularly the millimeter-wave (mmWave) spectrum ranging from 24 GHz to 100 GHz. These bands offer abundant unused bandwidth, allowing data transmission rates that far exceed those possible with traditional sub-6 GHz frequencies used in 4G and earlier technologies. For example, 5G promises data speeds in the order of multiple gigabits per second, which can support ultra-high-definition video streaming, real-time gaming, and immersive virtual reality experiences.

However, mmWave frequencies also present challenges. Due to their short wavelengths, mmWave signals experience significant free-space path loss, poor penetration through obstacles like walls or foliage, and susceptibility to atmospheric absorption. To overcome these issues, 5G networks rely on dense deployment of base

stations and advanced antenna technologies, including massive MIMO and beamforming.

Massive MIMO involves using large arrays of antennas at base stations to serve multiple users simultaneously by spatially multiplexing signals. This technique improves spectral efficiency and link reliability but requires antennas that are compact, efficient, and capable of precise beam steering. Beamforming further focuses the transmitted energy in specific directions, compensating for the high path loss at mmWave frequencies and enhancing signal quality.

In addition to physical layer improvements, 5G architecture incorporates network slicing, edge computing, and ultra-reliable low-latency communication (URLLC), enabling diverse applications from smart factories to autonomous vehicles. The complex and demanding nature of 5G systems underscores the critical importance of innovative antenna design.

Antenna technology for 5G is rapidly evolving, with research focusing on miniaturized arrays, reconfigurable designs, metamaterial integration, and intelligent prediction models. These advancements aim to achieve antennas that can deliver high gain, wide bandwidth, low mutual coupling, and consistent performance in real-world environments.

In this thesis, the focus is on the design and analysis of a compact 4-port MIMO antenna optimized for the 28 GHz mmWave band—a key frequency designated for 5G globally. By combining state-of-the-art antenna design practices with machine learning-based directivity prediction, this work contributes to the ongoing development of efficient antenna solutions vital for the success of 5G technology.

1.3 Importance of Antenna Design in 5G Wireless Systems

In next-generation wireless communication systems like 5G and beyond, antenna design takes on a central role in determining overall system performance. The antenna is the gateway through which wireless signals are transmitted and received, so its characteristics directly impact data rates, coverage, reliability, and energy efficiency.

Unlike earlier wireless generations, 5G operates at higher frequency bands, especially in the millimeter-wave (mmWave) spectrum. These frequencies offer wider bandwidths but come with challenges such as higher propagation losses and reduced

penetration through obstacles. To overcome these issues, antennas must be carefully designed to maximize gain and directivity, focusing energy efficiently toward desired directions while minimizing losses.

Key roles of antenna design in next-generation systems include:

1.3.1 Enhancing Signal Strength and Coverage:

Well-designed antennas with high gain and focused radiation patterns improve signal strength at the receiver, extending coverage areas even at mmWave frequencies where signals tend to attenuate rapidly.

1.3.2 Enabling MIMO and Beamforming:

MIMO technology relies on multiple antenna elements to transmit and receive parallel data streams. The antenna design must ensure low mutual coupling and high isolation between elements to maintain independent communication channels. Similarly, beamforming techniques require antennas capable of steering beams dynamically to track users and avoid interference.

1.3.3 Supporting Compact and Integrated Designs:

Modern wireless devices, from smartphones to IoT sensors, require antennas that are small, low-profile, and integrable into compact hardware while maintaining excellent performance. Efficient antenna design balances miniaturization with performance metrics like bandwidth and radiation efficiency.

1.3.4 Improving Spectral and Energy Efficiency:

Directional antennas reduce interference by limiting radiation to specific spatial regions, enabling better frequency reuse and higher spectral efficiency. Moreover, efficient antennas consume less power, contributing to greener wireless networks.

1.3.5 Adapting to Environmental Challenges:

Advanced antenna designs consider real-world deployment conditions, such as multipath propagation, user mobility, and physical obstructions. Robust antenna designs ensure consistent performance despite these challenges.

1.4 Basic Microstrip Patch Antenna

1.4.1 Structure and Operation of Microstrip Patch Antennas

Microstrip patch antennas (MSAs) are among the most widely used types of antennas in modern wireless communication due to their low profile, ease of fabrication, and suitability for integration with planar circuits. Structurally, an MSA consists of three primary components:

Radiating Patch: This is a metallic conductor, typically shaped as a rectangle or other geometries, placed on top of a dielectric substrate. The patch acts as the primary radiating element of the antenna.

Dielectric Substrate: A layer of insulating material that separates the patch from the ground plane. The substrate's thickness and dielectric constant significantly affect antenna characteristics such as bandwidth and efficiency.

Ground Plane: A conducting layer positioned beneath the dielectric substrate, serving as a reference plane for the antenna and aiding in directing the radiation pattern.

The operation of an MSA relies on the excitation of electromagnetic waves in the patch and its surrounding dielectric environment. When the antenna is fed (commonly through microstrip line, coaxial probe, or aperture coupling), the patch resonates at a frequency determined primarily by its dimensions and the substrate properties. At resonance, the patch radiates electromagnetic energy due to fringing fields formed at the edges of the patch, which act like radiating slots. The effective length and width of the patch dictate the resonant frequency, and can be approximated using established design formulas. The fringing fields are critical in antenna radiation and are influenced by the substrate thickness and dielectric constant. The antenna's performance parameters, such as gain, directivity, bandwidth, and polarization, depend on the careful design and optimization of these structural components.

1.4.2 Advantages and Disadvantages

Microstrip patch antennas offer several advantages, making them highly popular in a variety of applications:

Advantages:

Low Profile and Lightweight: MSAs are thin and flat, making them easy to mount on flat surfaces and integrate into compact devices.

Cost-Effective Manufacturing: The planar structure allows mass production using printed circuit board (PCB) fabrication techniques, reducing costs.

Ease of Integration: MSAs can be easily integrated with microwave integrated circuits (MICs), making them suitable for modern wireless systems.

Mechanical Robustness: The antennas are relatively robust and can be made durable for various environments.

Flexible Polarization: MSAs can be designed to operate with linear, circular, or elliptical polarization as per system requirements.

Multiband Operation: With appropriate design modifications, MSAs can operate over multiple frequency bands.

Despite these advantages, MSAs also have some limitations:

Disadvantages:

Narrow Bandwidth: Traditional MSAs suffer from limited bandwidth, typically a few percentage points of the center frequency, which restricts their ability to handle wideband signals.

Low Gain: The gain of a single patch antenna is modest, often requiring antenna arrays for higher gain applications.

Conduction and Dielectric Losses: Losses in the metal and substrate materials can reduce radiation efficiency.

Surface Wave Excitation: The substrate can support surface waves that cause unwanted radiation and reduce antenna efficiency.

Sensitivity to Fabrication Tolerances: Small variations in manufacturing can affect antenna performance, especially at higher frequencies.

1.4.3 Applications in Modern Communication Systems

Microstrip patch antennas have found widespread application in modern communication systems, owing to their versatility and ease of integration. Some notable applications include:

Mobile and Satellite Communications: MSAs are extensively used in mobile handsets and satellite transceivers due to their compact size and acceptable performance.

Radar Systems: Their planar design and ease of forming arrays make MSAs suitable for radar applications, including automotive and defense radars.

Global Positioning Systems (GPS): MSAs are commonly employed in GPS receivers, offering lightweight and low-profile antenna solutions.

Wireless Local Area Networks (WLAN) and Wi-Fi: MSAs are integral to WLAN devices, providing compact antennas capable of operating in 2.4 GHz and 5 GHz bands.

5G and mmWave Communications: Advanced MSAs, including those integrated with metasurfaces and forming MIMO arrays, are critical in 5G base stations and user devices for high-frequency operations.

Internet of Things (IoT): The compactness and low cost of MSAs make them ideal for embedding in numerous IoT devices requiring reliable wireless connectivity.

1.5 Multiple-Input Multiple-Output (MIMO) Antennas

1.5.1 Fundamentals of MIMO Technology

Multiple-Input Multiple-Output (MIMO) technology is a revolutionary approach in wireless communications that uses multiple antennas at both the transmitter and receiver ends. Unlike traditional single-antenna systems, MIMO leverages spatial diversity by simultaneously transmitting and receiving multiple data streams over different antennas. This approach takes advantage of multipath propagation — the phenomenon where transmitted signals reach the receiver via multiple paths due to reflection, diffraction, and scattering — turning what was once a challenge into a benefit.

In MIMO systems, the use of multiple antennas allows the wireless channel to support several independent data streams simultaneously, increasing the effective data throughput without requiring additional bandwidth or transmit power. This significantly improves link reliability and spectral efficiency, making MIMO a key enabling technology for modern wireless standards such as 4G LTE and 5G.

The core idea is to exploit spatial multiplexing, where independent signals are transmitted over separate antennas, and spatial diversity, where multiple antennas help combat fading and interference by providing multiple versions of the signal. Signal processing algorithms at the receiver combine these multiple inputs to reconstruct the transmitted information with improved quality.

1.5.2 Advantages of MIMO in Wireless Systems

The adoption of MIMO technology in wireless communication systems offers several critical advantages:

Increased Data Rates: By transmitting multiple independent data streams in parallel, MIMO effectively multiplies the channel capacity, enabling higher data rates for end users.

Improved Link Reliability: Spatial diversity helps mitigate the effects of multipath fading, leading to more reliable communication links even in challenging environments such as urban areas with many obstacles.

Enhanced Spectral Efficiency: MIMO maximizes the use of available frequency spectrum, an increasingly scarce and valuable resource, without requiring additional bandwidth.

Better Coverage and Capacity: With antenna arrays, beamforming can focus energy toward specific users, improving signal strength and reducing interference to others, which boosts overall network capacity.

Energy Efficiency: By improving link quality, MIMO allows for reduced transmit power while maintaining performance, contributing to energy savings in wireless networks.

1.5.3 Challenges in MIMO Antenna Design

Despite its advantages, designing MIMO antennas introduces several challenges that must be addressed to realize the full potential of the technology:

Mutual Coupling: When multiple antennas are placed close together, electromagnetic coupling between them can occur, leading to undesired signal interference and degradation of system performance. High isolation between antenna elements is crucial to minimize this effect.

Correlation Between Antenna Elements: The degree to which signals received by different antennas are correlated affects diversity gain. Designing antennas that produce low envelope correlation coefficient (ECC) is essential to maximize spatial diversity benefits.

Compactness vs. Performance Trade-offs: To fit multiple antennas into small mobile devices or compact base stations, antenna elements must be closely spaced, which often increases mutual coupling and correlation, reducing efficiency.

Bandwidth and Frequency Stability: Ensuring that all antenna elements operate effectively across the required bandwidth without performance degradation is challenging, especially at millimeter-wave frequencies.

Fabrication and Integration Complexity: Designing and manufacturing multi-element antenna arrays with precise control over geometry and electrical parameters require advanced fabrication techniques and careful integration with other device components.

Maintaining Efficiency and Gain: As the number of elements increases, maintaining high radiation efficiency and gain across all elements becomes difficult due to increased losses and mutual interactions.

1.6 Key Antenna Parameters and Performance Metrics

1.6.1 S-Parameters and Return Loss

S-parameters, or scattering parameters, describe how radio frequency (RF) signals are reflected and transmitted through an antenna or network. Among these, the most

critical parameter for antenna performance is S11, which indicates the reflection coefficient at the antenna input port. This parameter quantifies the amount of power that is reflected back toward the source due to impedance mismatch. Ideally, for efficient antenna operation, most of the input power should be radiated rather than reflected, which corresponds to a low S11 value. Typically, an S11 less than -10 dB is considered acceptable, signifying that over 90% of the power is successfully transmitted into the antenna. Return loss, expressed in decibels (dB), is closely related and represents the power lost due to reflections. A high return loss means better impedance matching and minimal reflected power, ensuring efficient radiation. Figure 1.1 illustrates the concept of S-parameters and the reflection and transmission coefficients in an antenna system.

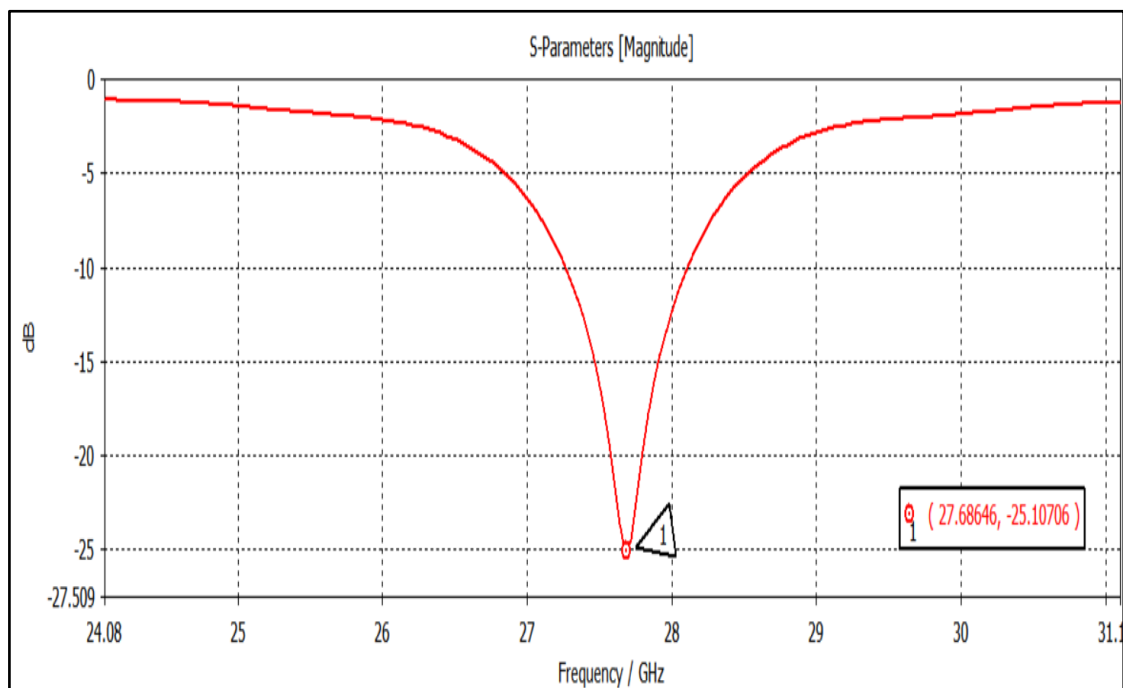


Figure 1.1 return loss (S11) vs Frequency graph of 5G antenna

1.6.2 Radiation Pattern

The radiation pattern of an antenna describes how the power is distributed spatially around the antenna. It provides a visual and quantitative depiction of the antenna's ability to radiate energy in various directions. The main lobe of the radiation pattern

shows the direction in which the antenna emits the highest power, which is the desired direction for signal transmission or reception. Aside from the main lobe, side lobes and back lobes represent unintended radiation in other directions, which can cause interference and reduce system efficiency. These patterns are often measured and plotted in specific planes, such as the electric field plane (E-plane) and the magnetic field plane (H-plane). Figure 1.2 shows a typical antenna radiation pattern

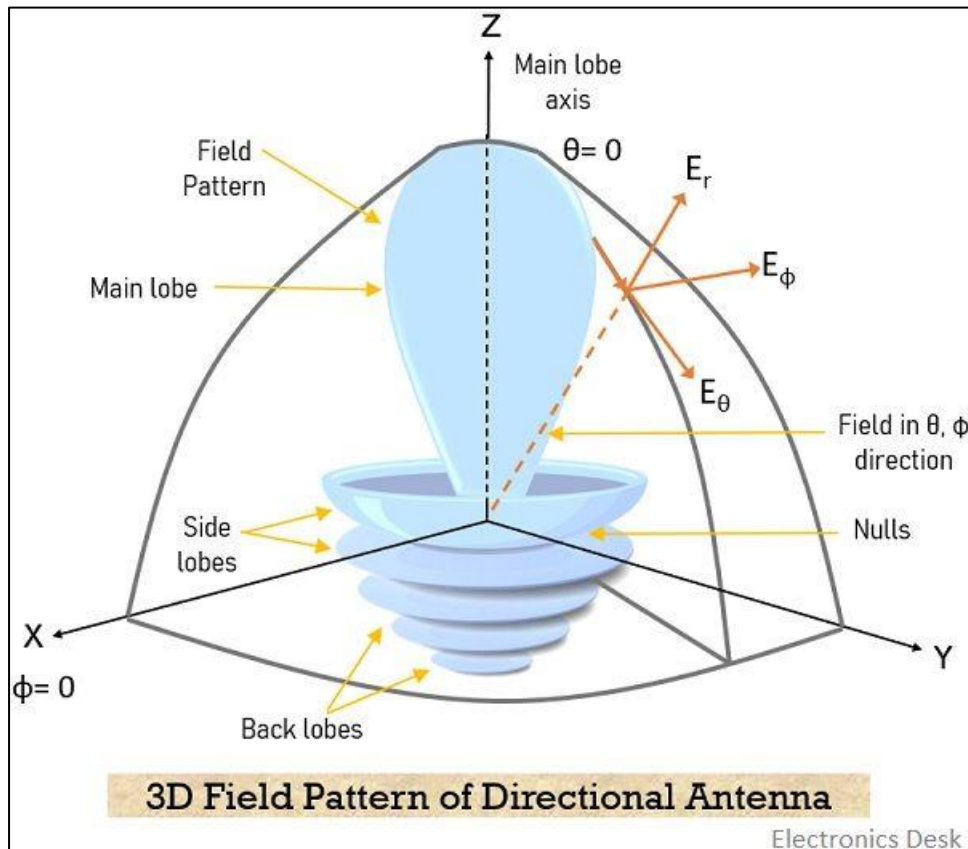


Figure 1.2 3D Radiation pattern of antenna

1.6.3 Directivity

Directivity is a measure that quantifies how concentrated an antenna's radiated energy is in a particular direction compared to an ideal isotropic radiator, which radiates energy uniformly in all directions. It is defined as the ratio of the maximum radiation intensity in the direction of the main lobe to the average radiation intensity over the entire sphere. A higher directivity value indicates that the antenna concentrates its energy more tightly in a specific direction, making it suitable for long-distance

communication and reducing interference. Directivity is a dimensionless ratio but is commonly expressed in decibels relative to an isotropic radiator (dBi). Figure 1.3 illustrates the concept of antenna directivity by comparing focused beam radiation to an isotropic source.

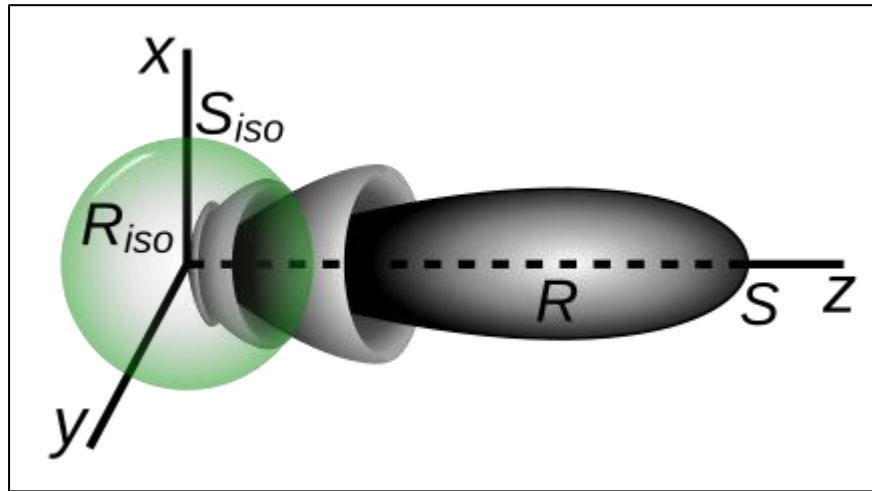


Figure 1.3 Directivity & Gain of antenna

1.6.4 Gain

Gain represents the ability of an antenna to direct or concentrate radiated power in a particular direction, considering both the directivity and the antenna's efficiency. Unlike directivity, gain accounts for losses inside the antenna caused by conductor resistance, dielectric materials, and other factors. It is measured relative to an isotropic antenna and expressed in dBi. High gain antennas produce stronger signals in their main radiation direction, improving signal quality and communication range. Figure 1.3 presents a visual explanation of antenna gain and its relationship to radiation patterns. The diagram illustrates the concept of antenna gain by comparing a directive antenna to an isotropic antenna. The axes represent the power density measured in watts per square meter (W/m^2) at a certain distance from the antenna.

The directive antenna has a radiation pattern labeled RRR, which concentrates its power in specific directions. At its strongest point along the z-axis, it radiates a maximum power density of $S \text{ W/m}^2$. Because the directive antenna focuses its energy within a narrow angle around the z-axis, it achieves a higher signal strength in that direction compared to the isotropic antenna, leading to a gain value greater than one. This gain quantifies how much more effectively the directive antenna radiates

energy in its main lobe compared to the idealized isotropic source.

Finally, gain and directivity remain fundamental performance metrics. Directivity defines how well an antenna focuses energy in a particular direction relative to an isotropic source:

$$D = \frac{4\pi U_{\max}}{P_{\text{tot}}}$$

where $U_{\max}$ is the maximum radiation intensity and P_{tot} is the total radiated power. Gain incorporates antenna efficiency η into this measure:

$$G = \eta \times D$$

1.6.5 Radiation Efficiency

Radiation efficiency defines the effectiveness of an antenna in converting the input power into radiated electromagnetic waves. It is expressed as the ratio of radiated power to the total power supplied to the antenna. Inefficiencies occur due to losses in the antenna's metallic conductors, dielectric substrate, and other materials, as well as through mismatch losses and surface waves. High radiation efficiency, close to 100%, is desirable as it ensures that most of the input power is effectively radiated into free space, enhancing communication performance.

1.6.6 Voltage Standing Wave Ratio (VSWR)

Voltage Standing Wave Ratio (VSWR) is a metric used to assess how well the antenna impedance matches the characteristic impedance of the feeding transmission line, typically standardized at 50 ohms. VSWR is defined as the ratio of the maximum to minimum voltage of the standing wave formed on the transmission line due to reflections caused by impedance mismatch. A VSWR of 1:1 represents perfect matching, meaning there is no reflected power and all energy is delivered to the antenna. Higher VSWR values indicate greater mismatch and increased reflected power, leading to inefficiencies and potential damage to the transmitter.

$$VSWR = \frac{V_{max}}{V_{min}}$$

1.6.7 Input Impedance and Impedance Matching

Input impedance is the complex ratio of voltage to current at the antenna terminals, consisting of resistive and reactive components. Proper impedance matching between the antenna and transmission line ensures maximum power transfer and minimizes reflections. The resistive part accounts for the radiated power and losses, while the reactive part relates to stored energy in the antenna structure. If the antenna input impedance does not match the transmission line impedance, reflected power increases, reducing efficiency. Impedance matching can be achieved through careful antenna design or by adding external matching networks.

1.6.8 Polarization

Polarization defines the orientation and time variation of the electric field vector of the radiated wave. It is a critical factor affecting signal reception and interference. The most common types of polarization are linear, circular, and elliptical. Linear polarization occurs when the electric field oscillates in a single plane, either horizontal or vertical. Circular polarization is formed when two orthogonal linear components of equal magnitude are out of phase by 90 degrees, causing the electric field to rotate in a circle over time. Elliptical polarization is a general case where the electric field describes an ellipse. Matching polarization between the transmitting and receiving antennas is essential for efficient communication, as mismatched polarization can result in signal loss.

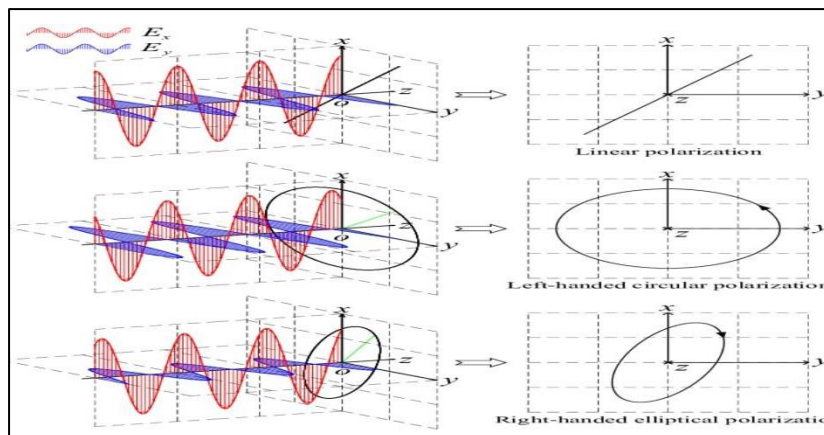


Figure 1.4 Showing different types of polarization

1.7 MIMO Antenna Parameters

1.7.1 S-Parameters (Reflection and Transmission Coefficients)

S-parameters describe how RF signals behave within a multi-port antenna system, focusing on reflection and transmission between antenna elements. The reflection coefficient (S_{11}) measures the proportion of power reflected back from each antenna port, with lower values indicating better impedance matching and efficient radiation. Transmission coefficients (such as S_{21} , S_{31}) quantify power coupled between different antenna elements, with low transmission values indicating high isolation essential for minimizing interference in MIMO systems. This behavior can be visualized in a multi-port antenna system graph plotting these coefficients versus frequency, highlighting efficient matching and isolation. The reflection coefficient is defined as:

$$S_{11} = \frac{V_{\text{reflected}}}{V_{\text{incident}}}$$

Also

$$S_{ij} = \frac{V_{\text{input at port } i}}{V_{\text{output at port } j}}$$

1.7.2 Isolation and Mutual Coupling

Isolation quantifies the electromagnetic separation between antenna elements placed close to each other. High isolation reduces unwanted signal leakage and interference, which can distort radiation patterns and degrade system performance. Isolation is often expressed in decibels as:

$$\text{Isolation} = -20\log_{10} |S_{21}|$$

where S_{21} represents the transmission coefficient between antennas.

1.7.3 Envelope Correlation Coefficient (ECC)

The envelope correlation coefficient (ECC) measures the similarity or correlation between signals received or transmitted by different antenna elements in a MIMO system. A low ECC value close to zero indicates strong spatial diversity, which improves communication reliability, while high ECC values suggest correlated signals that reduce diversity benefits. ECC can be approximated using antenna radiation patterns

$$\text{ECC} \approx \frac{|S_{11}^* S_{12} + S_{21}^* S_{22}|^2}{(1 - (|S_{11}|^2 + |S_{21}|^2))(1 - (|S_{22}|^2 + |S_{12}|^2))}$$

where S_{11}, S_{22} , are the reflection coefficients at ports 1 and 2 respectively, and S_{12}, S_{21} are the transmission coefficients (mutual coupling) between the two ports. The asterisk S^* denotes the complex conjugate. This formula is widely used for ECC calculation when radiation efficiencies are high.

1.7.4 Diversity Gain (DG)

Diversity gain (DG) quantifies the improvement in signal quality and robustness achieved by using multiple antenna elements compared to a single antenna. It reflects the system's capability to mitigate fading and multipath effects. DG is calculated directly from ECC as:

$$\text{DG} = 10\sqrt{(1 - (\text{ECC})^2)}$$

1.7.5 Channel Capacity Loss (CCL)

Channel capacity loss (CCL) represents the degradation in theoretical maximum data throughput due to correlation and mutual coupling between MIMO channels.

$$\text{CCL} = -\log_2 \det(\mathbf{PR})$$

Where PR is power radiated.

1.7.6 Gain and Directivity

Finally, gain and directivity remain fundamental performance metrics. Directivity defines how well an antenna focuses energy in a particular direction relative to an isotropic source:

Typical mmWave MIMO antennas exhibit gains between 7 and 9 dBi with directivities near or above 9 dBi, offering strong, focused radiation suitable for high-capacity wireless links.

$$D = \frac{4\pi U_{\max}}{P_{\text{tot}}}$$

where $U_{\max}$ is the maximum radiation intensity and P_{tot} is the total radiated power. Gain incorporates antenna efficiency η into this measure:

$$G = \eta \times D$$

Where G is gain, η is efficiency and D is directivity of antenna.

1.8 Machine Learning in Antenna Design and Optimization

With the growing complexity of antenna systems, especially for modern technologies like 5G, traditional design and simulation methods can become time-consuming and computationally expensive. This is where Machine Learning (ML) comes in as a powerful tool to revolutionize antenna design and optimization.

1.8.1 Introduction to ML Techniques in Antenna Design

Machine learning is a branch of artificial intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed for every specific task. In antenna design, ML techniques analyze vast amounts of simulation or experimental data to recognize patterns and relationships between antenna parameters and their performance metrics, such as gain or directivity.

1.8.2 Benefits of ML in Predicting Antenna Performance

Using ML models allows designers to quickly predict antenna behavior for new

configurations without running lengthy electromagnetic simulations each time. This speeds up the design process, reduces computational costs, and can even discover innovative designs by exploring parameter spaces that are otherwise difficult to investigate manually. ML helps to automate the optimization process, making antenna design more efficient and accessible.

1.8.3 Overview of Regression Models Relevant to Antenna Parameter Prediction

Among the ML techniques, regression models are commonly used for predicting continuous antenna performance values. These include Decision Trees, Random Forests, Extreme Gradient Boosting (XGB), and Multivariate Adaptive Regression Splines (MARS). These models learn from training data—sets of antenna designs with known performance—and then predict outcomes for new designs with impressive accuracy. Choosing the right model depends on the complexity of the problem, available data, and desired prediction speed.

1.9 Problem Statement and Research Gaps

Despite the advances in MIMO antenna designs for 5G and other high-frequency applications, several challenges and limitations remain.

1.9.1 Limitations of Existing MIMO Antenna Designs

Many current MIMO antenna systems struggle to maintain high gain, broad bandwidth, and excellent isolation simultaneously within compact sizes required by modern devices. Moreover, as frequencies increase into the millimeter-wave band, design tolerances become tighter, and mutual coupling effects between antenna elements grow, which degrade performance.

1.9.2 Challenges in Directivity Prediction and Gain Enhancement

Accurate prediction of antenna directivity and gain typically relies on time-intensive full-wave electromagnetic simulations. These simulations can be computationally expensive, especially when optimizing antennas with many variables. Additionally, enhancing gain and directivity without compromising other parameters, like isolation or bandwidth, remains a complex balancing act.

1.9.3 Gaps in Integrating ML Techniques for Antenna Optimization

While ML has shown promise in antenna design, its integration into MIMO antenna optimization—especially for directivity and gain prediction at mmWave frequencies—is still underexplored. Most studies focus on single-element antennas or lower frequencies, and there is a need for more research combining ML with advanced MIMO antenna designs tailored for 5G.

1.10 Aims and Objectives

The primary goal of this thesis is to address the above challenges by designing a compact and highly efficient 4-port MIMO antenna operating at 28 GHz for 5G applications, and to utilize machine learning techniques to predict and optimize antenna directivity and gain effectively.

1.10.1 Design and Fabrication of Compact 4-Port MIMO Antenna

Develop a compact, multi-element MIMO antenna using appropriate substrate materials and antenna geometries that ensure wide bandwidth, high gain, and strong isolation.

1.10.2 Performance Evaluation: Gain, Directivity, Isolation, Efficiency

Conduct comprehensive simulation and measurement to evaluate key antenna parameters including gain, directivity, port isolation, envelope correlation coefficient, and radiation efficiency to verify the antenna's suitability for 5G applications.

1.10.3 Application of Machine Learning for Directivity Prediction

Implement and compare various machine learning regression models to predict antenna directivity and gain from design parameters, thereby reducing the need for extensive simulations.

1.10.4 Improvement of Antenna Parameters Using ML-Assisted Methods

Use the insights gained from ML predictions to guide design modifications that improve antenna performance metrics efficiently and effectively.

1.11 Scope of the Thesis

This thesis focuses on the design, simulation, fabrication, and measurement of a compact 4-port MIMO antenna specifically for operation in the 28 GHz mmWave frequency band—a key spectrum allocated globally for 5G communications. It integrates machine learning techniques to predict antenna performance, enabling faster and more efficient antenna design cycles.

While this study covers a detailed design and performance analysis, it is limited to the 28 GHz frequency band and specific antenna configurations on Rogers RT/duroid substrates. Deep learning or other advanced ML models requiring very large datasets are beyond the current scope. The thesis also assumes ideal laboratory conditions for measurement and simulation, with limited exploration of real-world environmental effects.

1.12 Thesis Organization

This thesis is structured into several chapters to methodically develop the research. Chapter 1 introduces the background, motivation, problem statement, aims, and scope of the thesis. Chapter 2 provides a review of existing literature on 5G MIMO antenna designs, the application of machine learning in antenna engineering, and relevant performance metrics. Chapter 3 details the methodology, including the antenna design process, simulation environment, and development of the machine learning model. Chapter 4 presents the results obtained from simulations, physical measurements, and ML predictions, along with a comprehensive discussion on antenna performance. Finally, Chapter 5 concludes the thesis with a summary of the key findings, the research contributions made, and recommendations for future work.

1.13 Conclusion

To sum up, this chapter has highlighted the important role that advanced 4-port MIMO antennas play in meeting the demands for faster data speeds, better reliability, and wider coverage in 5G mmWave networks. It also discussed the key challenges faced in antenna design, especially when trying to achieve high gain and good isolation within a small form factor. The chapter introduced how machine learning is becoming a valuable tool to predict and improve antenna performance more efficiently. The goals and scope of this thesis were laid out to tackle these issues by

combining innovative antenna design with machine learning techniques. With this groundwork in place, the next chapter will take a closer look at the existing research, reviewing past developments in MIMO antenna design, performance evaluation, and the use of machine learning in antenna engineering. This review will help pinpoint the gaps that this thesis aims to address and shape the approach taken moving forward.

Chapter 02

Literature Review

2.1 Introduction

The rapid evolution of wireless communication technologies has transformed how information is transmitted and received worldwide. With the emergence of fifth-generation (5G) networks, the demand for higher data rates, lower latency, and massive device connectivity has increased exponentially. To meet these demands, 5G utilizes millimeter-wave (mmWave) frequency bands, particularly near 28 GHz, which provide much wider bandwidths compared to previous generations. However, operating at these higher frequencies presents significant challenges, including increased signal attenuation, limited propagation distances, and sensitivity to environmental obstacles.

Antenna technology plays a vital role in addressing these challenges. Compact, efficient, and high-performance antennas are essential components in 5G infrastructure and user devices. Specifically, Multiple-Input Multiple-Output (MIMO) antenna systems, which employ multiple antenna elements for simultaneous transmission and reception, have become indispensable. MIMO systems enhance spectral efficiency and reliability by exploiting spatial diversity and multiplexing, which is critical in the complex propagation environments of mmWave frequencies. This chapter reviews the current landscape of mmWave antenna design for 5G applications, with a particular focus on compact 4-port MIMO antenna systems. It discusses the design principles, challenges, and performance metrics that govern these antennas. Furthermore, it explores the integration of modern computational techniques, such as machine learning, to accelerate and improve antenna design and optimization. This literature review forms the basis for the methodology and research contributions presented in subsequent chapters.

2.2 Review of mmWave Antennas for 5G

The adoption of mmWave frequencies for 5G offers a promising solution to the growing demand for faster wireless communication. The 28 GHz band, in particular, has been allocated globally due to its ability to support wide bandwidth channels necessary for high data throughput. However, mmWave propagation suffers from high free-space path loss, atmospheric absorption, and poor penetration through common obstacles like buildings and foliage, which necessitates specialized antenna designs capable of overcoming these drawbacks.

Researchers have extensively studied antennas that can operate efficiently in this challenging environment. Haque et al. [1] proposed a compact 4-port MIMO antenna designed on a Rogers RT/duroid 5880 substrate that achieves a wide impedance bandwidth of 3.7 GHz and a high gain of approximately 8.9 dB at 28 GHz. The antenna also exhibits excellent isolation greater than 29 dB and radiation efficiency near 98%, highlighting its suitability for mmWave 5G applications. This design demonstrates that careful substrate selection, optimized patch geometry, and precise feeding techniques can produce antennas that meet stringent 5G performance requirements.

Parallel investigations by Rana and Rahman [2] similarly underline the importance of substrate material properties and dimensional accuracy in achieving efficient mmWave antenna performance. Their work emphasizes how dielectric constant and substrate thickness affect bandwidth, gain, and efficiency. Such considerations are critical because minor deviations can significantly impact antenna behavior at high frequencies.

The increasing interest in 5G has spurred innovations beyond traditional microstrip patch antennas. Advanced methods, including the integration of metamaterials and metasurfaces, have been explored to enhance antenna bandwidth, gain, and isolation without enlarging antenna size. The use of array configurations and corporate feed networks further enhances the effective aperture, contributing to higher gain and better beam steering capabilities, essential for compensating mmWave propagation losses.

Collectively, these studies confirm the feasibility and necessity of compact, efficient MIMO antennas designed for the mmWave 28 GHz band. They also lay the groundwork for the integration of computational intelligence techniques, such as machine learning, to further refine antenna design and performance prediction topics elaborated later in this chapter.

2.3 Design Techniques for Compact and High-Gain Antennas

Designing antennas for the mmWave 28 GHz band requires a careful balance between compactness and high gain to counteract the increased path loss at higher frequencies. Microstrip patch antennas are often preferred for their low profile, ease of fabrication, and integration capabilities with planar circuits. However, conventional microstrip patches suffer from inherent limitations such as narrow bandwidth and moderate gain, which restrict their effectiveness in 5G applications [2,6]. To address these limitations, researchers have introduced various design enhancements.

One common technique is the incorporation of slots or parasitic elements into the patch, which effectively increases the electrical length of the antenna without enlarging its physical size, thereby enhancing bandwidth and gain [14]. For instance, Hakeem et al. [8] demonstrated how etching specific slot shapes into microstrip patches can broaden impedance bandwidth and improve radiation characteristics at mmWave frequencies. Corporate feeding networks are also employed in array designs to improve power distribution and impedance matching, leading to increased gain and bandwidth as shown in the work by Muhammad and Idris [4]. Their research illustrates that corporate feed designs in multi-element arrays can achieve better performance metrics compared to single-element antennas.

The choice of substrate material plays a significant role as well. Rogers RT/duroid 5880, with its low dielectric constant and loss tangent, is widely used for mmWave antenna designs due to its minimal dielectric losses and stable electrical properties at high frequencies [1,2]. Optimizing substrate thickness and dielectric properties can significantly influence antenna efficiency and bandwidth.

Furthermore, the integration of metamaterials and metasurfaces is an emerging design trend that offers the possibility to manipulate electromagnetic waves in unconventional ways to improve antenna performance metrics without increasing size [1,14]. By carefully engineering these artificial materials, designers can achieve enhanced gain, improved isolation, and reduced mutual coupling, all critical for compact high-gain antenna systems suitable for 5G.

2.4 State-of-the-Art MIMO Antennas for 28 GHz Band

Multiple-Input Multiple-Output (MIMO) antenna systems leverage spatial multiplexing and diversity to enhance data throughput and link reliability in 5G networks. The 28 GHz mmWave band poses unique challenges for MIMO antenna design, such as increased mutual coupling due to element proximity and the need for compact arrays.

Haque et al. [1] designed a compact 4-port MIMO antenna system operating at 28 GHz, achieving a wide operational bandwidth of 3.7 GHz, gain of up to 8.94 dB, and port isolation exceeding 29 dB. This design effectively balances compactness and performance, utilizing orthogonal placement of antenna elements and optimized feed networks to minimize coupling. The antenna also demonstrated a very low envelope correlation coefficient (ECC) of less than 0.00005 and a diversity gain (DG) close to 10 dB, indicating excellent diversity and channel capacity characteristics.

Complementary research by Tiwari et al. [3] optimized a 2x2 MIMO patch antenna array using machine learning techniques to explore the design space efficiently and improve bandwidth and gain. This study confirmed that intelligent algorithms could facilitate the design of high-performance MIMO arrays while reducing development time.

Other studies, such as those by Sohel Rana et al. [11], focus on machine learning for MIMO antenna parameter prediction, reflecting a broader trend toward integrating computational intelligence in antenna design. Furthermore, recent works by Sudais [9] and investigations into reconfigurable antennas show promising advances in dynamic adaptation of antenna properties to suit varying 5G communication scenarios.

Collectively, these contributions demonstrate that compact, efficient 4-port MIMO antennas capable of meeting stringent 5G mmWave requirements are achievable using a combination of careful physical design and advanced computational tools, including machine learning, which is discussed further in this thesis.

2.5 Optimization Methods in Antenna Design

2.5.1 Traditional Parametric Sweeps and Simulation

The conventional antenna design process heavily relies on parametric sweeps performed within full-wave electromagnetic simulators such as CST Microwave Studio. Designers systematically vary parameters like patch dimensions, feed positions, and substrate properties to observe their impact on antenna performance. Although this approach is straightforward and widely used, it becomes computationally expensive and time-consuming when dealing with complex multi-element MIMO antennas that have numerous design variables. As the design space grows exponentially, exhaustive sweeps are often impractical [3][4].

2.5.2 Introduction to Machine Learning Techniques

Machine learning (ML) has recently become a transformative tool in antenna design. By training on simulation data generated from CST, ML models learn complex nonlinear relationships between antenna design parameters and performance metrics. Once trained, these models predict antenna characteristics quickly without needing further full-wave simulation runs. Haque et al. [1] utilized ML regression models to accurately predict directivity of a 4-port MIMO antenna, significantly accelerating the design cycle and enabling real-time design decisions.

2.5.3 Supervised Learning and Regression Models

Supervised learning algorithms, particularly regression models such as Decision Trees, Random Forests, and Multivariate Adaptive Regression Splines (MARS), have proven effective in antenna performance prediction. These models are trained using datasets generated by CST simulations, enabling them to estimate antenna gain, directivity, bandwidth, and return loss for new antenna geometries with high accuracy. Haque et al. [1] found Random Forest regression to be particularly successful in predicting directivity, while Rana et al. [11] showed polynomial regression models could reliably predict bandwidth and VSWR, thereby reducing prototyping needs.

2.5.4 Challenges in Data Collection and Model Training

Despite the benefits, ML-based antenna design faces challenges such as the need for extensive, high-quality datasets covering diverse antenna configurations. Gathering such datasets requires considerable simulation time and expertise, especially at mmWave frequencies where experimental validation is complex and costly [1][11]. Moreover, ML models must generalize well to novel antenna designs and remain robust to manufacturing tolerances and environmental effects, which current models often do not fully address. Overcoming these challenges is critical for wider adoption of ML in practical antenna engineering.

2.6 Application of Machine Learning in Antenna Design and Prediction

As antenna systems, particularly at mmWave frequencies, grow in complexity, traditional trial-and-error and exhaustive simulation-based design approaches become increasingly inefficient. Machine learning (ML) techniques have emerged as powerful tools that can learn from existing data to predict antenna performance metrics rapidly, significantly speeding up the design process.

ML models leverage large datasets generated through CST simulations or experimental measurements to uncover hidden relationships between antenna design parameters (such as patch dimensions, feed position, and substrate properties) and performance outcomes (like gain, bandwidth, and directivity). This approach eliminates the need for repeated full-wave electromagnetic simulations for every new design iteration, thus conserving computational resources and accelerating innovation [1][11].

Haque et al. [1] demonstrated the use of various regression models, including Random Forests, Multivariate Adaptive Regression Splines (MARS), and Decision Trees, to predict the directivity of a compact 4-port MIMO antenna designed for 28 GHz applications. Their study showed that Random Forest regression achieved superior accuracy and generalization capabilities, reducing prediction errors while requiring significantly less computational time than traditional simulations.

Similarly, Sohail Rana et al. [11] employed polynomial regression models to predict key antenna parameters such as bandwidth, return loss, and Voltage Standing Wave Ratio (VSWR). Their findings confirmed that ML-assisted models could reliably

forecast antenna performance with minimal training data, demonstrating the feasibility of integrating ML into practical antenna design workflows.

Moreover, machine learning approaches have been used not only for prediction but also for optimization. Tiwari et al. [3] combined ML with heuristic optimization to enhance antenna gain and bandwidth, exemplifying how data-driven algorithms can guide antenna parameter selection efficiently.

However, while ML promises faster and more adaptive antenna design processes, it also introduces challenges, including the need for sufficiently large and representative training datasets and the risk of overfitting models to limited data. Additionally, incorporating real-world effects such as manufacturing tolerances and environmental variability into ML models remains an active area of research [1][11].

2.7 Research Gaps and Motivation for Current Study

While significant progress has been made in the design and optimization of mmWave MIMO antennas for 5G, several critical gaps remain that motivate the current research.

2.7.1 MIMO Design Difficulty

Designing compact MIMO antennas that meet the rigorous demands of 5G mmWave systems is inherently challenging. Achieving high gain, wide bandwidth, and excellent isolation simultaneously in a small form factor requires intricate antenna element arrangements and sophisticated feeding structures [1][2]. The mutual coupling effects between closely spaced antenna elements further complicate the design process, often leading to performance degradation if not carefully managed [1][5]. Consequently, realizing a MIMO antenna that balances all these parameters remains a non-trivial task requiring advanced design strategies.

2.7.2 Complexity of Design and Computational Cost

Traditional design approaches rely heavily on full-wave electromagnetic simulations using software like CST Microwave Studio. These simulations become increasingly time-consuming and computationally expensive as the number of antenna elements and design variables grow [3][4]. Multi-parameter optimization via parametric sweeps or heuristic algorithms can be prohibitive for rapid design iteration, slowing down development cycles. Even with optimization algorithms and evolutionary methods,

the computational cost remains a bottleneck, especially for mmWave MIMO arrays where small changes in geometry have significant effects on performance [3].

2.7.3 Dataset Compiling for Machine Learning Models

Machine learning offers a promising solution to accelerate antenna design; however, it depends critically on the availability of large, high-quality datasets for training models [1][11]. Generating these datasets involves running extensive simulations or performing numerous measurements, which can be costly and time-intensive, particularly at mmWave frequencies where fabrication and testing are more complex [1].

2.8 Summary

This chapter provided a comprehensive overview of the current research landscape surrounding mmWave MIMO antennas for 5G applications. It highlighted the critical importance of designing compact, high-gain, and wideband antennas that can operate efficiently at the 28 GHz frequency band. Various design techniques and advanced feeding methods were discussed alongside performance metrics such as bandwidth, isolation, and radiation efficiency that govern antenna effectiveness. The chapter also explored how modern optimization strategies, particularly machine learning, are revolutionizing antenna design by enabling rapid, accurate prediction of antenna parameters and significantly reducing computational costs. Despite these advancements, significant challenges remain, most notably the complexities of multi-element MIMO antenna design, the high computational burden of traditional optimization methods, and the scarcity of extensive, high-quality datasets needed for effective machine learning model training. These gaps form the foundation of the current research motivation. Building on this background, Chapter 3 details the methodology and design process adopted in this thesis. It elaborates on the step-by-step antenna design, simulation environment using CST Microwave Studio, and the development and application of machine learning regression models for directivity prediction. This integrated approach aims to overcome the identified challenges by combining innovative antenna architecture with data-driven performance prediction, setting the stage for efficient and effective 5G antenna development.

Chapter 3

Methodology and Design

3.1 Design Specifications and Requirements for MIMO Antenna

The design of antennas for 5G millimeter wave (mmWave) applications involves a number of critical performance requirements due to the high-frequency operation and the demand for compactness, efficiency, and reliability. In the context of the 5G mmWave band particularly around the 28 GHz frequency the antenna must meet stringent performance metrics to ensure effective integration into modern communication systems. This section outlines the key design specifications and requirements that guided the development of the proposed 4-port MIMO antenna.

3.1.1 Operational Frequency and Bandwidth:

For 5G mmWave applications, the 28 GHz band is one of the most widely adopted frequency ranges due to its support for high-speed, low-latency data transmission. Therefore, the antenna was designed to operate in the frequency range of 24 GHz to 32 GHz, providing a wide impedance bandwidth of approximately 2 GHz (26.6-28.6 GHz). This broad bandwidth ensures that the antenna can accommodate various 5G services and protocols without signal degradation or dropouts.

3.1.2 Gain and Radiation Efficiency:

High antenna gain is essential in mmWave communication to compensate for high path losses at such elevated frequencies. For this design, a minimum target gain of 8.5 dB was set, with simulated results reaching up to 8.94 dB. Additionally, radiation efficiency above 95% was deemed necessary to minimize power loss and enhance overall system performance. The final design achieved an efficiency of 98.4%, which aligns well with commercial standards.

3.1.3 Isolation and MIMO Performance Metrics:

Since the antenna system employs a 4-port MIMO configuration, mutual coupling between the ports needed to be minimized. An isolation level of greater than 29 dB

between antenna elements was targeted to reduce interference and improve signal diversity. In parallel, MIMO performance metrics such as the Envelope Correlation Coefficient (ECC) and Diversity Gain (DG) were considered. An ECC value of less than 0.0007 and a DG of around 10 dB were specified to ensure low signal correlation and high diversity performance.

3.1.4 Antenna Footprint and Substrate Material:

Compactness is a key requirement for 5G hardware integration, especially for mobile or handheld devices. Therefore, the antenna was designed to occupy an area not exceeding $3.3\lambda_0 \times 3.3\lambda_0$. Rogers RT/duroid 5880 was chosen as the substrate due to its low dielectric constant ($\epsilon_r = 2.2$), low loss tangent ($\tan\delta = 0.0009$), and excellent high-frequency characteristics. The substrate thickness was selected to be 0.787 mm, balancing mechanical stability and electrical performance.

3.1.5 Impedance Matching and Fabrication Compatibility:

For seamless integration with other RF components and standard measurement systems, the antenna was designed with a 50-ohm impedance match. The physical structure including feedlines, slots, and patch dimensions—was optimized using parametric studies to ensure accurate resonance at 28 GHz while maintaining fabrication simplicity.

3.2 Substrate Material Selection and Its Impact on Performance

The substrate material plays a critical role in the performance of any microstrip antenna, particularly when operating at high frequencies such as those used in 5G millimeter wave (mmWave) applications. The choice of substrate affects key antenna parameters including impedance bandwidth, gain, efficiency, and overall radiation characteristics. For this project, Rogers RT/duroid 5880 was selected as the substrate due to its exceptional electrical and mechanical properties, which are well-suited for high-frequency applications like the 28 GHz 5G band.

3.2.1 Electrical Properties:

Rogers RT/duroid 5880 is known for its low dielectric constant ($\epsilon_r = 2.2$) and extremely low loss tangent ($\tan\delta = 0.0009$). A low dielectric constant allows for better radiation efficiency, which is crucial at mmWave frequencies where losses are

inherently higher due to atmospheric absorption and surface scattering. Additionally, the low loss tangent minimizes dielectric losses within the substrate, thereby enhancing the antenna's total radiation efficiency. In the case of this design, these material properties contributed to achieving a simulated efficiency of 98.4%, which is significantly high for compact MIMO antennas operating at such frequencies.

3.2.2 Thickness Consideration:

The substrate thickness used in this design was 0.787 mm. This value was chosen to strike a balance between mechanical stability and performance. Thicker substrates generally improve bandwidth but may lead to higher surface wave losses, while thinner substrates can limit bandwidth and reduce structural strength. The selected thickness offered an ideal compromise, supporting a broad bandwidth of 3.7 GHz while maintaining mechanical robustness during the fabrication and testing stages.

3.2.3 Impact on Antenna Size and Miniaturization:

Due to the relatively low dielectric constant, the guided wavelength within the substrate is longer. This slightly increases the physical size of the patch elements but significantly improves the impedance matching and radiation characteristics. For this reason, the antenna design achieved a compact layout ($3.3\lambda_0 \times 3.3\lambda_0$) while maintaining high isolation and minimal signal degradation between ports, which is crucial for MIMO performance.

3.2.4 Fabrication Suitability:

Another advantage of using Rogers RT/duroid 5880 is its compatibility with standard photolithographic PCB fabrication processes. Its dimensional stability, thermal resistance, and copper cladding properties make it suitable for precise etching, which is important at mmWave frequencies where even minor fabrication errors can lead to significant performance deviations.

3.2.5 Comparison with Other Materials:

Although alternative substrates like FR-4 are widely used due to their low cost, they are not suitable for high-frequency applications due to their higher loss tangent and dielectric constant. This results in higher losses, poor impedance matching, and lower efficiency, especially above 10 GHz. By contrast, Rogers RT/duroid 5880 maintains

consistent performance and is thus preferred for 5G antenna applications where performance cannot be compromised.

3.3 Theoretical Background and Design Equations for Microstrip Patch Antennas

Microstrip patch antennas have become a popular choice in modern wireless communication systems due to their low-profile structure, ease of fabrication, and compatibility with printed circuit technology. Their planar configuration and simple geometry make them especially suitable for high-frequency applications such as 5G millimeter wave (mmWave) systems. However, to achieve optimal performance at such high frequencies, a deep understanding of their theoretical operation and accurate application of design equations is essential.

A typical microstrip patch antenna consists of a radiating patch on one side of a dielectric substrate with a ground plane on the other. The most common shape for the radiating patch is rectangular due to its simplicity in analysis and fabrication. The patch operates as a resonant cavity where the radiation primarily occurs due to fringing fields at the edges of the patch.

The fundamental resonant frequency of a rectangular patch operating in the TM_{10} mode is given by:

$$f_r = \frac{c}{2L\sqrt{\epsilon_{eff}}}$$

Where:

- f_r is the resonant frequency
- C is the speed of light in free space
- L is the effective length of the patch
- ϵ_{eff} is the effective dielectric constant of the substrate

3.4 Single Element Antenna Design

The single element antenna serves as the foundational building block for the proposed 4-port MIMO antenna system. Before moving toward array or MIMO configurations, it is essential to ensure that the individual element performs well in terms of return

loss, gain, bandwidth, and radiation efficiency. The single patch antenna in this project was designed to operate at 28 GHz, which is a widely adopted frequency in 5G millimeter wave (mmWave) communication systems.

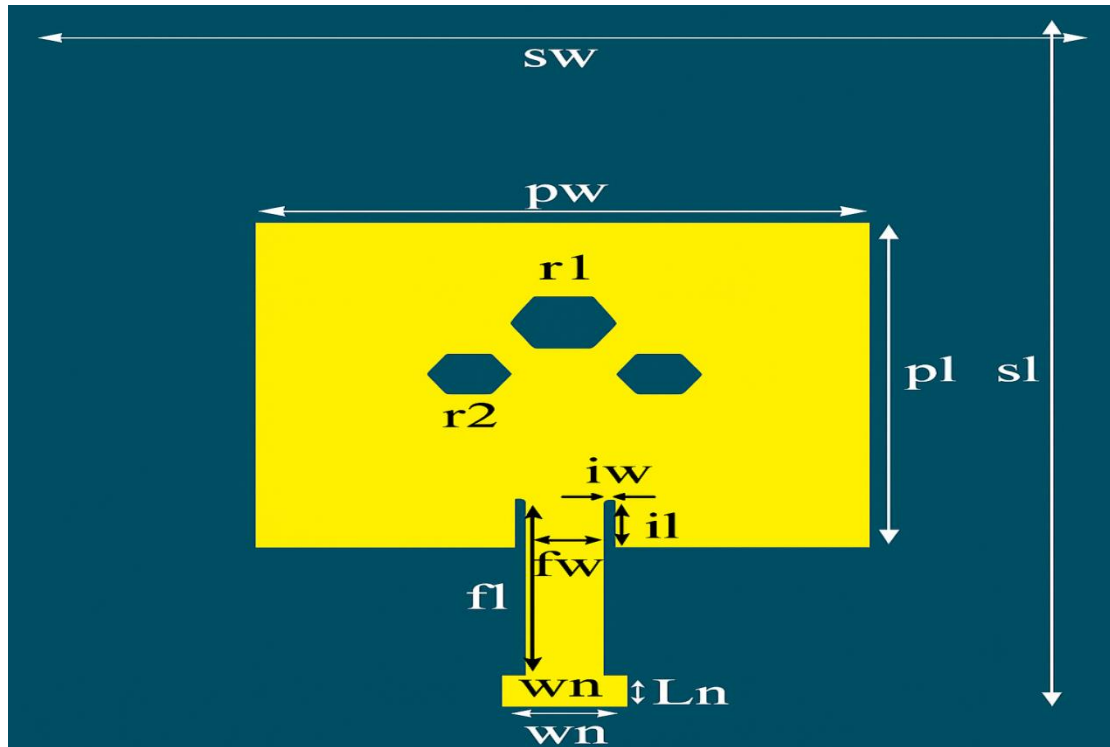


Figure 3.1 Single Element With dimensions

3.4.1 Design Diemensions

length of sustrate	sw	7mm
Width of substrate	sl	7mm
Length of patch	pl	2.993 mm
Width of patch	pw	4mm
Length of feed	fl	2.5mm
Width of feed	fw	0.7mm
Length of inset	il	0.787mm
Width of inset	wl	0.05mm
Height of sustrate	hs	0.787mm

Height of ground	hg	0.035mm
height of patch	(hp	0.035mm

Initial Design Considerations:

The design began with calculating the basic dimensions of a rectangular microstrip patch antenna using the theoretical equations described in the previous section. Based on the substrate properties (Rogers RT/duroid 5880, $\epsilon_r = 2.2$, height = 0.787 mm), the patch length and width were computed to achieve resonance at 28 GHz. These values served as the starting point for modeling the antenna in CST Microwave Studio.

• Design Stages and Modifications:

The development of the single element went through multiple design iterations. Initially, a conventional rectangular patch with a 50-ohm microstrip feedline was used. However, to enhance the bandwidth and achieve better return loss, structural modifications were introduced in subsequent stages. These included:

- Hexagonal slots etched into the patch to shift and broaden the resonant frequency range.
- Feedline insets and width adjustments to fine-tune impedance matching.
- Optimization of patch and substrate dimensions based on simulation feedback.

The antenna was gradually improved over four key stages:

1. **Stage 1:** Basic rectangular patch with a feedline and a top-centered hexagonal slot. Resonance was observed at 27.5 GHz with a bandwidth of 1.8 GHz and gain of 6.5 dB.
2. **Stage 2:** Addition of a second hexagonal slot improved return loss, shifted resonance to 27.5 GHz, and increased gain to 7 dB.
3. **Stage 3:** A third hexagonal slot enhanced bandwidth to 1.89 GHz and improved return loss further.
4. **Stage 4:** Feedline width modifications successfully brought resonance to exactly 27.5 GHz, achieving a 2 GHz bandwidth and gain of 7.5 dB.

Simulation Results and Observations:

The final version of the single element antenna exhibited a reflection coefficient (S_{11}) below -10 dB across the targeted frequency band, indicating good impedance matching. The gain stabilized around 7.5 dB, and the radiation pattern showed desirable characteristics for use in a MIMO configuration. The simulated surface current distribution at 27.5 GHz confirmed strong excitation around the feedline and patch edges, validating the antenna's efficiency and resonance behavior.

3.4.1 Patch Geometry and Feeding Mechanism

The initial geometry of the patch was chosen to be rectangular due to its well-understood radiation behavior and ease of fabrication. The patch dimensions—length and width—were determined based on the target resonant frequency of 28 GHz, considering the dielectric constant and thickness of the substrate (Rogers RT/duroid 5880, $\epsilon_r = 2.2$, $h = 0.787$ mm). The patch width was selected to optimize the radiation efficiency and bandwidth, while the patch length was calculated to ensure resonance at the desired frequency using standard microstrip antenna equations.

The feeding mechanism employed was a microstrip line feed, which is commonly used due to its planar structure and ease of integration with the patch. A 50-ohm microstrip line was designed to match the input impedance of the antenna and ensure minimal reflection. The feedline was connected directly to the patch, and a small inset was added to improve impedance matching. This feeding approach provided a balanced trade-off between design simplicity and effective excitation of the radiating element.

3.4.2 Slot Introduction for Bandwidth Enhancement

One of the key challenges in mmWave antenna design is achieving wide bandwidth without increasing the size or complexity excessively. To address this, slots were introduced into the patch as a method of modifying the current distribution and generating additional resonant paths, thereby enhancing the bandwidth.

A hexagonal slot was first etched into the top center of the rectangular patch. This modification shifted the resonant frequency and broadened the operational bandwidth by introducing an additional current path. In the second stage of design, an additional hexagonal slot was added at the lower right side of the patch, further tuning

the resonance characteristics and improving return loss. A third hexagonal slot was introduced at the lower left corner of the patch in the third design stage. This final slot further enhanced bandwidth and improved the symmetry of the radiation pattern.

The incorporation of slots not only improved the bandwidth from 1.8 GHz to 2 GHz but also helped in fine-tuning the resonant frequency closer to the 28 GHz target. The surface current distribution confirmed that the slots played a significant role in achieving these performance improvements by concentrating the current flow along critical paths.

3.4.3 Feedline Modifications and Optimization

After achieving satisfactory results from slot loading, the focus shifted toward optimizing the feedline to further refine impedance matching and ensure that the antenna resonated exactly at 28 GHz. Initially, a standard 50-ohm feedline was used; however, simulation results indicated minor mismatches and suboptimal return loss performance.

To address this, the feedline width and its transition to the patch were carefully adjusted. The width near the junction of the feedline and patch was increased slightly to control the current density and improve the matching condition. Furthermore, the inset length and width were also modified to fine-tune the feed point impedance. These refinements resulted in a well-matched antenna with a return loss below -30 dB at resonance, minimal reflection, and stable performance across the desired bandwidth. The optimized feedline not only ensured better impedance matching but also improved gain and enhanced the overall efficiency of the single-element antenna.

3.5 Parametric Studies for Single Element Antenna

Parametric analysis is a critical part of antenna design, especially at millimeter wave frequencies where even minor dimensional changes can significantly affect performance. In this section, a systematic study is conducted to observe how variations in key design parameters influence the antenna's resonant frequency, return loss, and bandwidth. The goal of these simulations was to identify the most sensitive parameters and optimize them for better matching, gain, and overall performance.

The parametric study focused on three major aspects: feed length, substrate width, and patch length. Each parameter was varied while keeping the rest of the design constant,

and the resulting changes in the reflection coefficient (S11) were observed using CST Microwave Studio.

3.5.1 Effect of Feed Length on Resonance

The feed length plays a crucial role in determining the impedance matching and resonance behavior of the antenna. An inappropriate feed length can shift the resonance away from the desired frequency, resulting in poor return loss and reduced radiation efficiency.

In the simulation, the feed length was varied from 2.3 mm to 2.5 mm, with the original design value set at 2.45 mm. The results showed that decreasing the feed length caused the resonance frequency to shift to the right (higher frequency), while increasing it led to a shift to the left (lower frequency). The optimal feed length of 2.45 mm provided the best matching at 28 GHz with a reflection coefficient lower than -30 dB. This confirms that slight variations in the feed structure can critically impact the antenna's performance at mmWave frequencies.

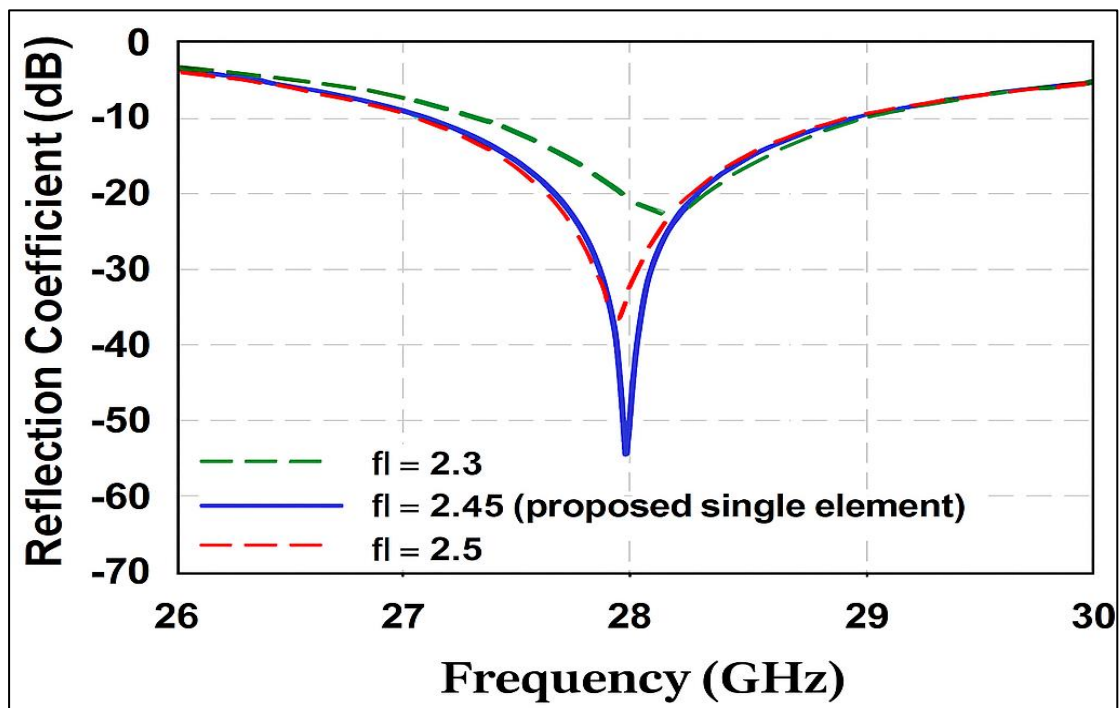


Fig 3.2 Effect of feed length on resonance frequency

3.5.2 Effect of Substrate Width

The width of the substrate also affects the effective dielectric constant and the propagation of surface waves, which in turn influence the resonant behavior of the patch. In this study, the substrate width was varied between 6 mm and 8 mm, with 7 mm being the baseline value.

The reflection coefficient plots indicated that changes in substrate width had a noticeable impact on the resonance frequency. A narrower substrate resulted in a leftward shift in resonance (lower frequency), whereas a wider substrate shifted the resonance to the right (higher frequency). Moreover, deviations from the optimal width led to poorer return loss values and narrower bandwidth. This highlights the importance of maintaining an appropriate substrate width to preserve impedance matching and minimize undesired effects such as surface wave losses and spurious radiation.

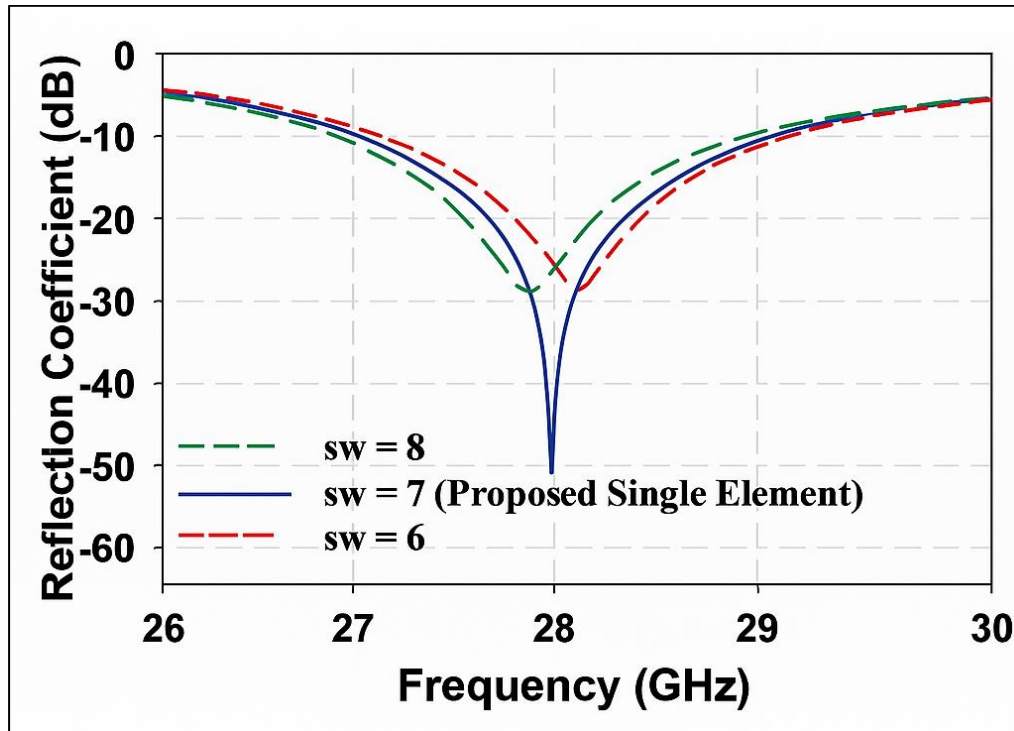


Figure 3.3 Effect of substrate width on resonance frequency

3.5.3 Effect of Patch Length

Among the most influential parameters in determining the resonant frequency of a patch antenna is the patch length. Since the patch behaves like a resonant cavity, even a small change in length can shift the resonance considerably.

In this analysis, the patch length was varied slightly above and below the optimized value of 2.993 mm. The simulation results showed a clear trend: increasing the patch length caused the resonance to move to a lower frequency, while decreasing it shifted the resonance to a higher frequency. When the patch length was set to the optimized value, the resonance occurred at 27.984 GHz with a strong reflection coefficient of -50 dB, indicating excellent matching and efficient operation.

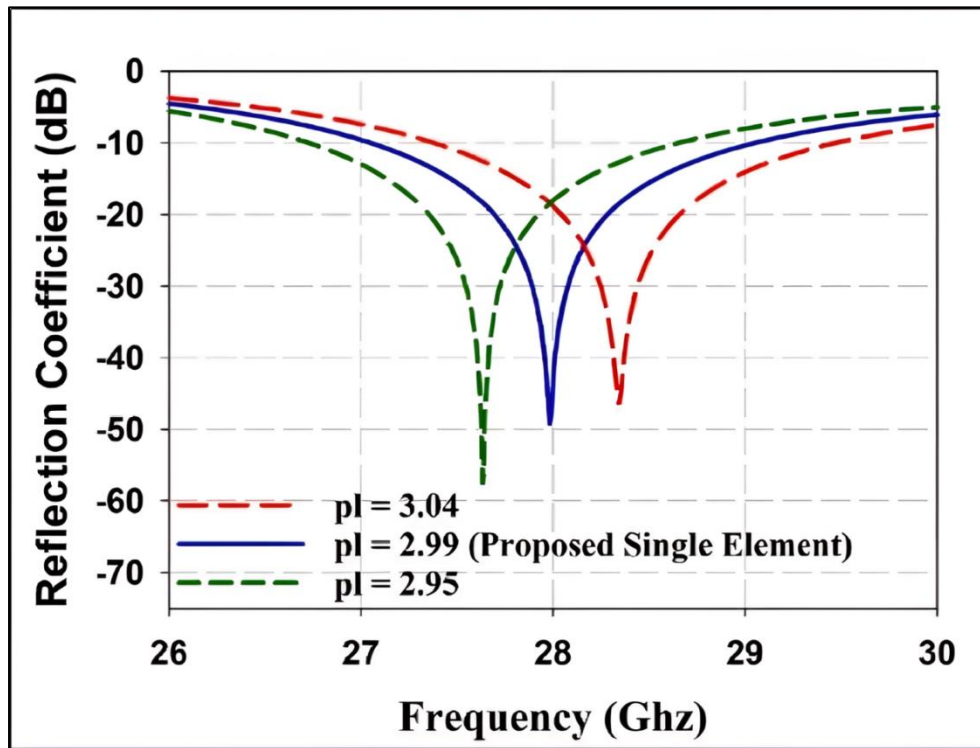


Figure 3.4 Effect of patch length on resonance frequency

3.6 Transition to Array Antenna

After achieving satisfactory performance from the single patch element, the next logical step in the antenna development process was to expand the design into an array configuration. Array antennas are known for offering improved gain, broader bandwidth, and enhanced radiation characteristics compared to single-element antennas. They are particularly beneficial in 5G mmWave applications where directional communication and beamforming are essential for overcoming path loss

and ensuring high data throughput.

In this project, a two-element array was designed as an intermediate step toward building a full 4-port MIMO antenna. This phase allowed for an investigation into mutual coupling, impedance matching in multi-element configurations, and the effect of element spacing on performance parameters.

3.6.1 2-Element Array Design and Optimization

The 2-element array was formed by duplicating the optimized single-element design and arranging the two patches side-by-side in a linear layout. A power divider structure was incorporated, where a 50-ohm main feedline branched into two 100-ohm feedlines connected to the individual patches. This branching network ensured equal power distribution and phase alignment, which are critical for uniform radiation in array systems.

During the initial implementation, each element retained the same dimensions and feeding configuration as in the single-element design. However, fine-tuning was necessary to address the new electromagnetic environment caused by mutual coupling. Several optimization iterations were conducted in CST Microwave Studio, focusing on:

- Feedline width and branching impedance
- Inset adjustments for each element
- Substrate size and edge clearances
- Overall array footprint

The final optimized array exhibited a gain improvement from 7.5 dB (single element) to around 8.7 dB. Bandwidth also increased, extending from 2 GHz to approximately 3.5 GHz (26 GHz – 29.5 GHz), showing that constructive interference and array resonance contributed to broader operational range. These results confirmed the successful transition from a standalone element to a compact, high-performance antenna array suitable for 5G applications.

3.6.2 Array Spacing and Its Effect on Gain and Bandwidth

One of the most important parameters in array design is the spacing between individual elements. This spacing directly influences the mutual coupling, gain, bandwidth, and overall radiation pattern of the antenna array. For this design, the center-to-center spacing was initially set to approximately one guided wavelength (λ_g) at 28 GHz, which corresponds to around 1.8 mm in free space.

A parametric sweep was conducted to analyze how variations in element spacing affect performance. The observations were as follows:

- **Tighter spacing ($< \lambda_g$):** Led to higher mutual coupling, deteriorating the isolation between elements and reducing overall gain. Additionally, bandwidth narrowed due to increased interactions between the radiating patches.
- **Wider spacing ($> \lambda_g$):** Improved isolation and enhanced gain, but introduced side lobes and disrupted the compactness of the design, which is undesirable for portable 5G devices.
- The optimal spacing of 1.8 mm provided a balanced trade-off between compactness, mutual coupling, and performance. At this spacing, the array achieved stable gain, wide bandwidth, and good impedance matching without excessive radiation in undesired directions.

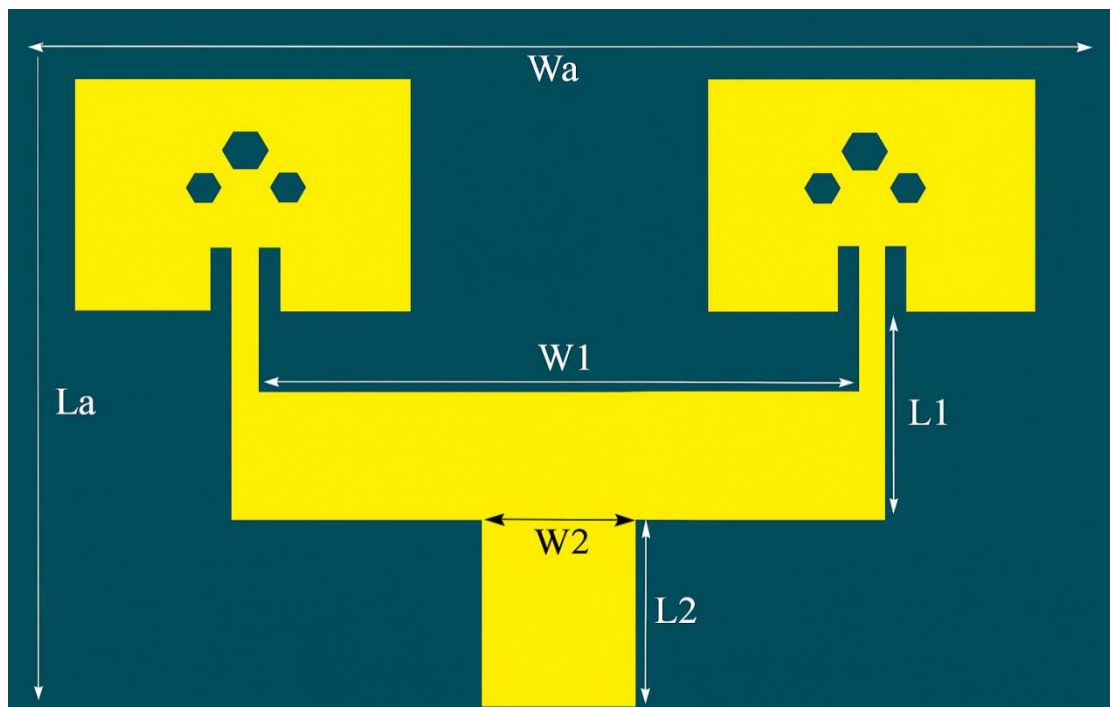


Figure 3.5 Illustrating 2-Element Array

3.7 Design and Optimization of 4-Port MIMO Antenna

Building upon the successful implementation of the single-element and 2-element array designs, the next stage in the antenna development process was the construction of a 4-port MIMO (Multiple Input Multiple Output) antenna. MIMO systems are essential in 5G wireless communication due to their ability to transmit and receive multiple data streams simultaneously, thereby increasing channel capacity and improving signal reliability. The 4-port configuration in this project was designed to ensure high gain, wide bandwidth, low envelope correlation, and high isolation between antenna elements.

This section discusses the design methodology, spatial arrangement of elements, techniques for minimizing mutual coupling, and the final optimized layout of the proposed MIMO antenna.

3.7.1 Element Arrangement and Spacing

The proposed 4-port MIMO antenna was designed by replicating the optimized single-element antenna and arranging the four elements in a square configuration. This layout was selected to achieve symmetry and uniform radiation in multiple directions, which is advantageous in MIMO systems operating in complex propagation environments.

Initially, the elements were placed with a gap of 17 mm from the center, resulting in close proximity and noticeable mutual coupling. To address this, the antenna positions were gradually shifted outward, and the inter-element distances were carefully adjusted. Two critical distances were defined:

- **D (Center-to-center spacing):** Set at 19 mm to maintain sufficient separation and prevent overlap of near fields.
- **G (Edge-to-edge spacing):** Optimized to 9.72 mm to reduce interference while keeping the design compact.
- This symmetrical spacing helped in maintaining consistent performance across all four elements and minimized undesired coupling effects.

3.7.2 Techniques for Mutual Coupling Reduction

In MIMO systems, mutual coupling between antenna elements can significantly degrade performance by introducing correlation and reducing diversity gain. To mitigate this, the following design strategies were implemented:

1. **Element Spacing Optimization:**

As described in Section 3.7.1, the physical spacing between antennas was increased from the initial layout to reduce electromagnetic interaction. This alone brought mutual coupling down from -22 dB to around -29.24 dB.

2. **Patch Rotation and Symmetry:**

Elements were positioned with rotational symmetry to ensure that surface currents and radiation patterns interfered minimally with each other.

3. **Use of Ground Plane and Isolation Gap:**

A continuous ground plane was maintained beneath the elements to control surface wave propagation. Additionally, the layout ensured that no feedline crossed into another element's active region, maintaining clean isolation paths.

4. **Simulation-Based Refinement:**

Using CST Microwave Studio, mutual coupling (S_{21} , S_{31} , S_{41}) was monitored and iteratively minimized during design optimization.

These techniques collectively helped achieve a high isolation level above 29 dB, which is well-suited for 5G MIMO requirements.

3.7.3 Final Optimized Layout and Dimensions

The final layout of the 4-port MIMO antenna achieved an excellent balance between compactness, performance, and manufacturability. The overall dimensions of the antenna were fixed at $35.58 \text{ mm} \times 35.58 \text{ mm}$, corresponding to approximately $3.3\lambda_0 \times 3.3\lambda_0$ at 28 GHz.

The complete detail of optimized design parameters is as follows (all in millimeters):

Table 3.1 Dimensions of final MIMO Antenna

Parameter	Description	Value (mm)
sw	Substrate width	35.58
sl	Substrate length	35.58
pw	Patch width	4
pl	Patch length	2.993
fw	Feed width	0.7
fl	Feed length	2.5
iw	Inset width	5.1
il	Inset length	0.84
$W_m \times L_m$	Board dimensions	35.58×35.58
D	Element spacing	19
G	Edge gap	9.72
t	Copper thickness	0.035
st	Substrate height	0.787

The final simulated results showed a return loss below -30 dB at 28 GHz, gain of approximately 8.9 dB, bandwidth of 3.7 GHz, and envelope correlation coefficient (ECC) less than 0.0007. These outcomes validate the antenna's capability to perform efficiently in a high-speed, high-density wireless communication environment.

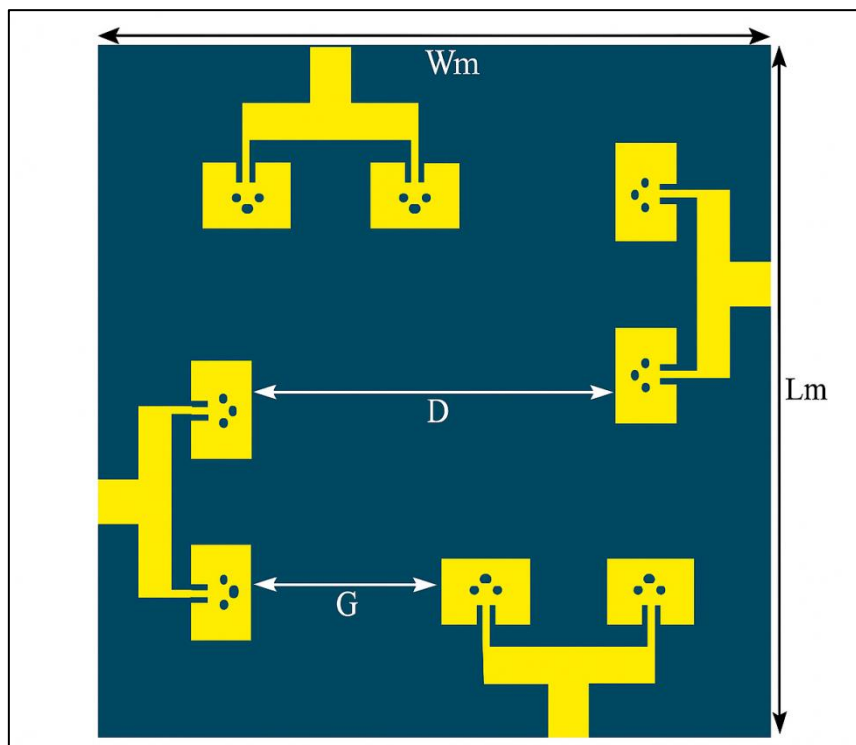


Figure 3.6 Showing 4-Port MIMO Antenna with mentioned dimensions

3.8 Simulation Environment and Tools

To design, analyze, and optimize the proposed 4-port MIMO antenna for 5G millimeter wave applications, advanced electromagnetic simulation tools were employed. Simulation plays a crucial role in antenna development, especially at high frequencies where experimental prototyping can be time-consuming and costly. The primary software used in this project was CST Microwave Studio, a full-wave 3D electromagnetic simulation tool based on the Finite Integration Technique (FIT). It allows for the accurate prediction of antenna parameters such as return loss, bandwidth, gain, radiation pattern, mutual coupling, and surface current distribution.

3.9 Machine Learning Approach for Directivity and Gain Prediction

In recent years, machine learning (ML) techniques have increasingly been applied to antenna design problems to reduce computational time, enhance prediction accuracy, and assist in intelligent optimization. In this project, machine learning was employed to predict key performance metrics specifically, directivity of the proposed MIMO antenna, based on various physical and design parameters.

By training regression models on data obtained from full-wave electromagnetic simulations, the ML approach served as a surrogate modeling tool. This enabled quick estimation of performance without the need for repeated CST simulations, thus saving valuable time during design iterations. The process included dataset generation, feature engineering, model selection, training, and performance evaluation.

3.9.1 Dataset Generation from Parametric Sweeps

To build a meaningful ML model, a high-quality dataset was generated through systematic parametric sweeps in CST Microwave Studio. Several geometric and material parameters of the single-element antenna were varied within realistic design limits to observe how they influence gain and directivity.

Over 120 simulation samples were created by varying parameters such as patch width, patch length, substrate width, feed length, and feed width. For each simulation run, CST was used to extract the corresponding gain and directivity at 28 GHz. The collected dataset was stored in CSV format for further processing and model training.

Each sample in the dataset represents one design configuration and its associated performance, making the dataset highly structured and suitable for supervised learning.

3.9.2 Input Features and Output Parameters

In supervised regression tasks, the quality of input features significantly affects the model's prediction accuracy. The input features for this task were selected based on their impact on antenna performance and included the following:

Input Features (Independent Variables):

- Patch Width (mm)
- Patch Length (mm)
- Substrate Width (mm)
- Feed Length (mm)
- Feed Width (mm)
- Gain (dB)

Output Parameters (Dependent Variables):

- Directivity (dBi).

3.9.3 Description of Regression Models

Five regression algorithms were employed and compared to identify the best-performing model for predicting gain and directivity:

1. Decision Tree Regressor:

A simple, tree-based model that splits the data into decision nodes based on feature values. It is interpretable but prone to overfitting on small datasets.

2. Random Forest Regressor:

An ensemble model that constructs multiple decision trees and averages their predictions. It offers improved accuracy and generalization over single decision trees.

3. XGBoost Regressor (XGB):

A gradient boosting-based model known for its high performance on structured

data. It handles bias-variance tradeoff efficiently and often outperforms traditional ensemble models.

4. Each model was implemented using Python's scikit-learn and xgboost libraries, and their performance was evaluated using standard regression metrics.

3.9.4 Training, Validation and Performance Metrics

A total of 320 iterations of data set is used for training machine learning models. The data set was split into training (70%) and testing (30%) sets. The models were trained on the training set and validated on the unseen test set to evaluate generalization. Cross-validation (5-fold) was also used to assess model robustness and prevent over fitting.

The following performance metrics were used:

- **Mean Absolute Error (MAE):** Measures the average magnitude of prediction errors.
- **Mean Squared Error (MSE):** Penalizes larger errors more severely than MAE.
- **Root Mean Squared Error (RMSE):** Square root of MSE; interpretable in the same unit as the output.
- **R² Score (Coefficient of Determination):** Indicates the proportion of variance explained by the model.
- **Variance Score:** Measures how well the model accounts for data variability.
- **Best Model:** XGB Gradient Boosting Regression
- **R²= 99.77%**
- **MAE= 0.05%**
- **RMSE=1%**
- **Efficency=99.79%**

The excellent R² score indicates that the model successfully captured the relationship between geometric parameters and antenna performance. This ML model proved to be a reliable tool for assisting in rapid design evaluations without repeated full-wave simulations.

3.10 Summary

Chapter 3 provided a comprehensive outline of the methodology and design process used to develop the 4-port MIMO antenna for 5G mmWave applications. The design specifications, including the operational frequency, bandwidth, gain, isolation, and efficiency requirements, were discussed in detail. The choice of substrate material—Rogers RT/duroid 5880—was justified based on its excellent performance at 28 GHz, providing high radiation efficiency and minimal dielectric losses. The single-element antenna design, including the use of slot modifications and feedline optimizations, was iteratively refined through parametric studies to meet the desired performance criteria.

Following the single-element design, the transition to a 4-port MIMO configuration was made, with careful consideration given to element arrangement, spacing, and techniques to reduce mutual coupling. Additionally, machine learning models were integrated into the design process to predict key performance metrics such as gain and directivity, reducing computational time and enabling faster iterations. The simulated results validated the design, achieving the targeted performance parameters.

The following chapter, Chapter 4: Results and Discussions, presents the outcomes of the simulations and measurements conducted to validate the design. It discusses the performance of the antenna in terms of gain, bandwidth, isolation, ECC, and the effectiveness of the machine learning models in predicting directivity. The results are analyzed and compared to the design goals, providing insights into the success of the proposed methodology and highlighting areas for further optimization.

Chapter 4: Results and Discussions

Introduction

This chapter presents and discusses the key results obtained from the simulation and validation of the proposed 4-port MIMO antenna for 5G applications. The antenna was designed, optimized, and simulated using CST Microwave Studio, with additional analysis performed using machine learning-based prediction models. The results cover bandwidth, VSWR, directivity, gain, return loss (S_{11}), mutual coupling, envelope correlation coefficient (ECC), diversity gain and radiation patterns for single element, 2-element array and 4-Port MIMO antenna. The results from machine learning models (XGB and SVM) are also discussed for directivity prediction on basis of different errors and R score (efficiency of model). This chapter not only summarizes the performance of the antenna but also compares it against design goals and existing literature to establish the novelty and effectiveness of the proposed methodology.

4.1 Single Element Antenna

Figure 4.1 shows a single element design with mentioned dimensions. It is designed using Rogers Rt duroid/5880 lossy substrate with width and length of 7mm and height of 0.787mm.

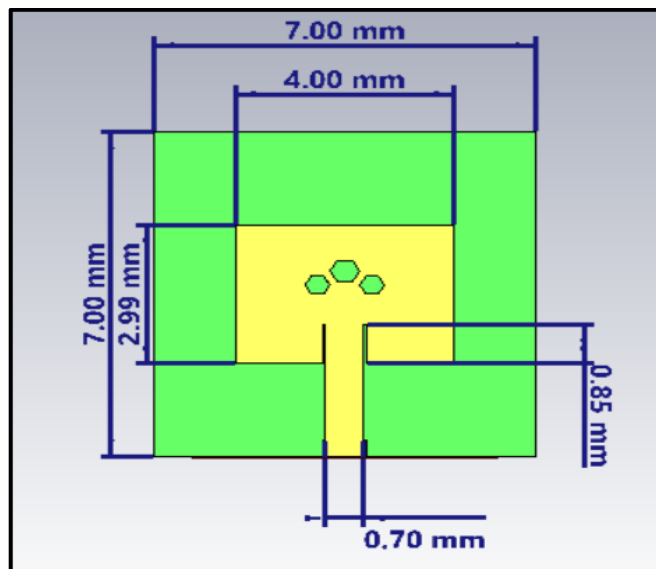


Figure 4.1 Single element Antenna design

4.2 Single Element Results

4.2.1 Reflection Coefficient (Return Loss) and Resonant Frequency

Return loss is power returned or reflected by antenna due to mismatching between transmission line and patch. Low value of S_{11} means less power is reflected and antenna is performing efficiently. Figure 4.2 shows graph of return loss and frequency of single element antenna. A low return loss of -64db is observed at resonant frequency 29.10 GHz

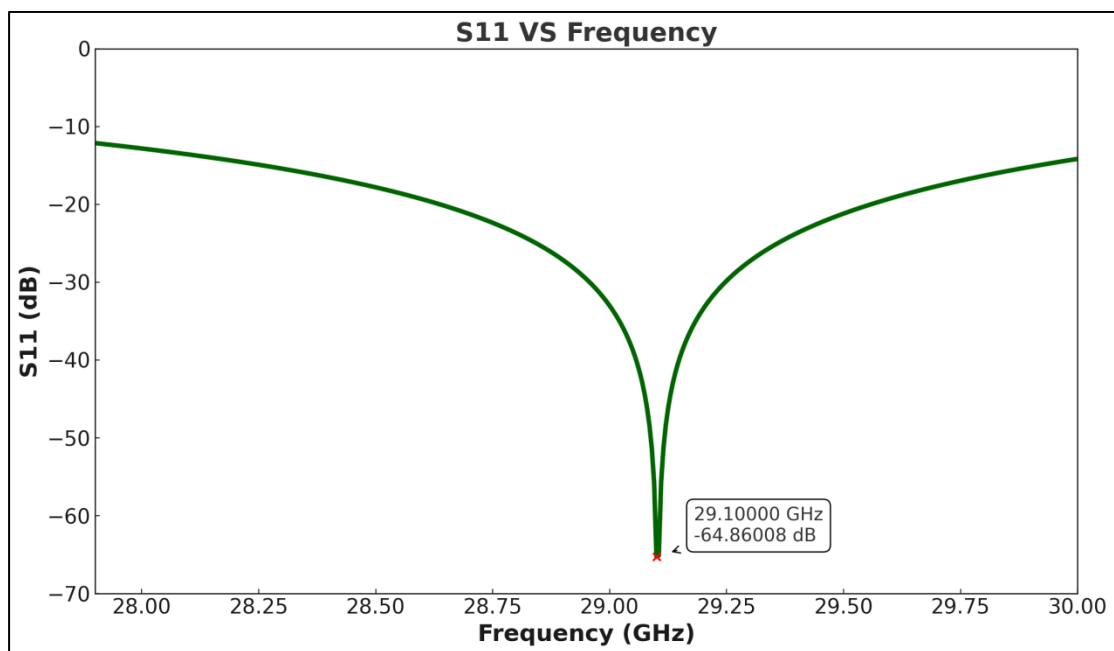


Figure 4.2 S11 and frequency Graph

4.2.2 Bandwidth

The results of the single-element microstrip patch antenna demonstrate a satisfactory performance within the desired frequency range for 5G mmWave applications. The antenna exhibits a bandwidth wide enough to cover the targeted 28 GHz band, with a return loss (S_{11}) below the typical threshold of -10 dB, indicating efficient impedance matching and minimal signal reflection across this range. Overall, the single-element antenna's bandwidth results validate its suitability as a fundamental building block for more complex MIMO configurations. In figure 4.2, we can observe that there is a high bandwidth of approximately 2GHz from 27.9 GHz to 30 GHz.

4.2.3 VSWR

The Voltage Standing Wave Ratio (VSWR) for the single-element antenna was measured to be approximately 1.00113 at 29.10 GHz, indicating an excellent impedance match at this frequency. A VSWR value this close to 1 means that nearly all the power delivered to the antenna is radiated efficiently with virtually no reflection back toward the source. This excellent matching ensures minimal signal loss and contributes to the antenna's overall effective performance. Such a low VSWR is highly desirable in 5G mmWave antennas, as it supports reliable signal transmission and maximizes the efficiency of the communication system. The detailed VSWR performance across the frequency band is illustrated in Figure 4.3, which clearly shows the antenna maintaining a VSWR well below the typical threshold of 2 throughout the operating range.

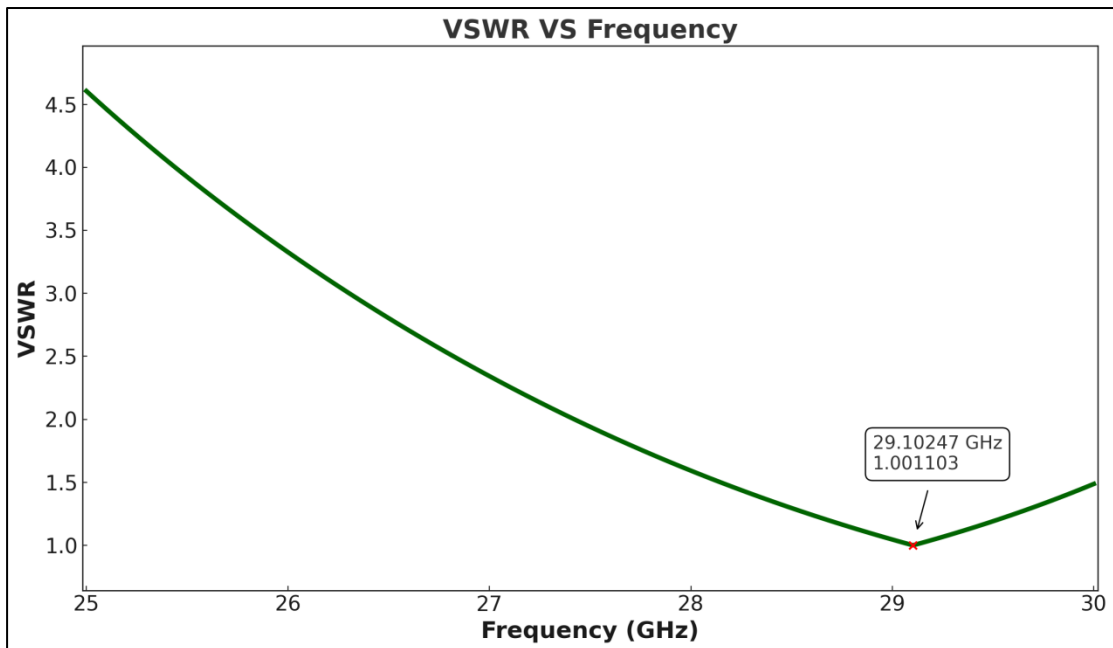


Figure 4.3 VSWR graph of single element antenna

4.2.4 Efficiency

The efficiency of the antenna is a crucial parameter that indicates how effectively the antenna converts the input power into radiated electromagnetic waves. A high-efficiency antenna ensures that minimal power is lost due to dielectric losses,

conductor losses, or surface wave generation. In our single-element antenna design, the efficiency was observed to be notably high across the operating frequency band, demonstrating that most of the input power is successfully radiated rather than dissipated as heat or trapped within the substrate. This relationship between the radiated power and the total input power is clearly illustrated in Figure 4.4, which shows the graph of radiated and simulated input power. The antenna efficiency η is calculated using the formula:

$$\eta = \frac{P_{\text{radiated}}}{P_{\text{input}}}$$

From the graph, at the operating frequency, the radiated power P_{radiated} is approximately 0.473 watts, while the input power P_{input} is about 0.5 watts. Substituting these values yields an efficiency of 94.6%. Such high radiation efficiency is essential for 5G mmWave applications, as it improves signal strength and conserves energy, ultimately enhancing device performance and battery life.

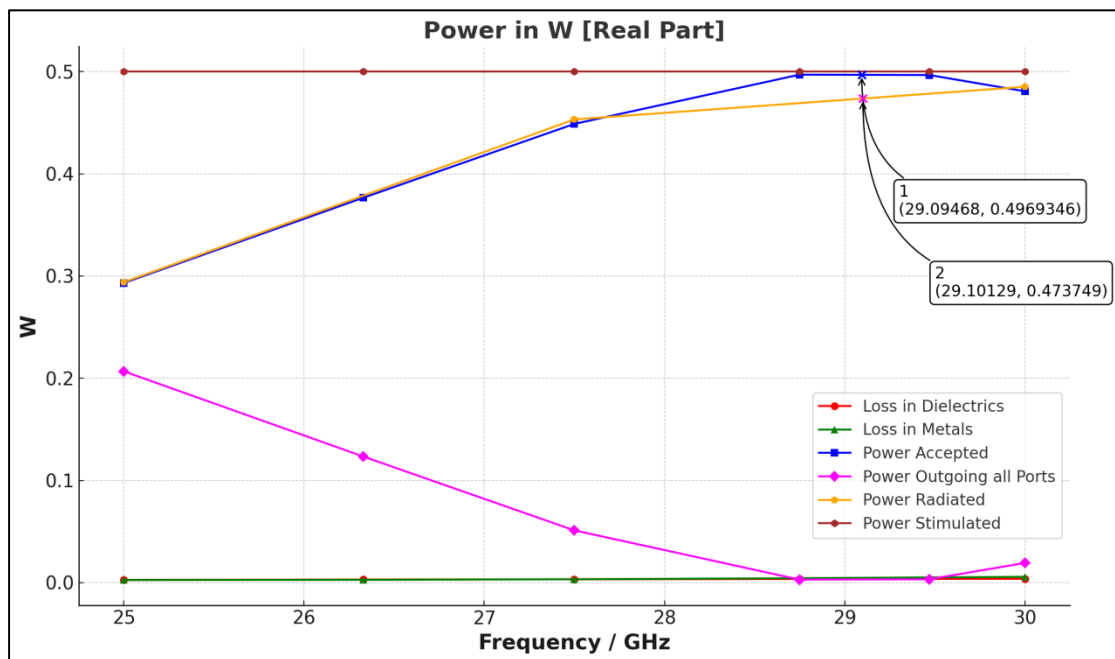


Figure 4.4 Power components vs. frequency showing accepted, radiated, stimulated powers and dielectric/metal losses.

4.2.5 Gain and Directivity

The single-element antenna demonstrates a gain of 7.25 dBi and a directivity of 7.21 dBi, indicating efficient concentration of radiated power in the desired direction. The main lobe magnitudes corresponding to these values are clearly shown in Figure 4.5 for gain and Figure 4.6 for directivity, respectively. The close agreement between gain and directivity values suggests minimal losses within the antenna structure, confirming its high radiation efficiency. This level of performance is well suited for 5G mmWave applications where focused radiation and efficient power transmission are crucial for maintaining signal strength and communication reliability in compact antenna designs.

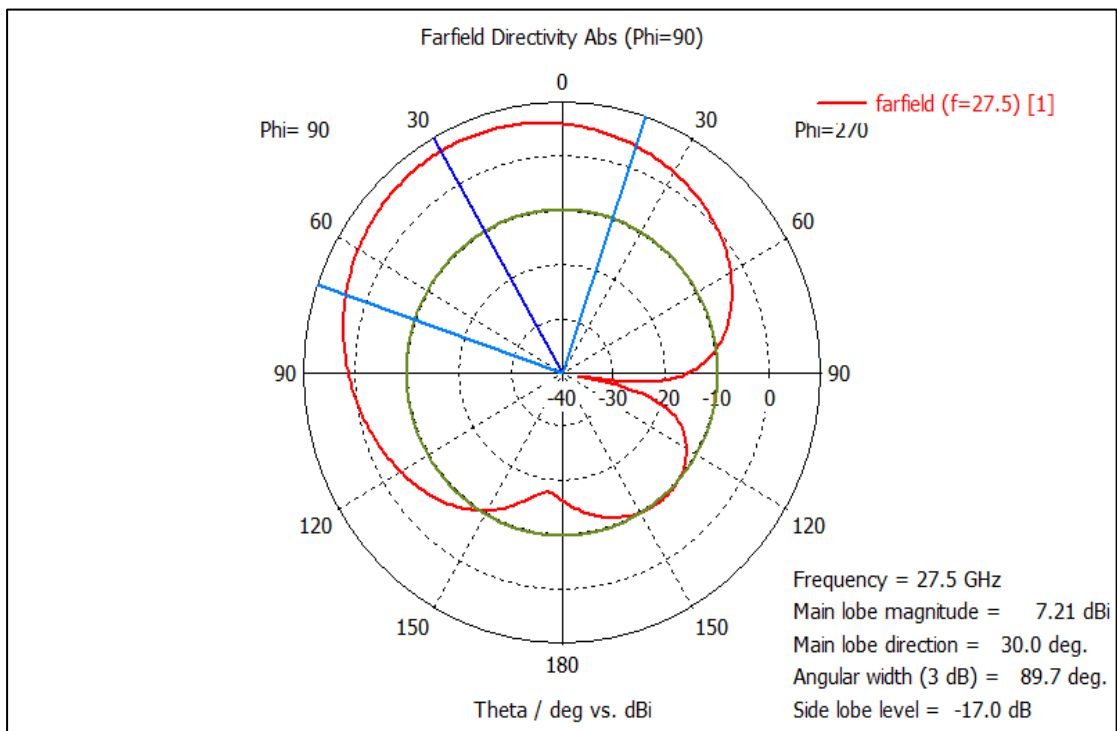


Figure 4.5 Directivity polar plot

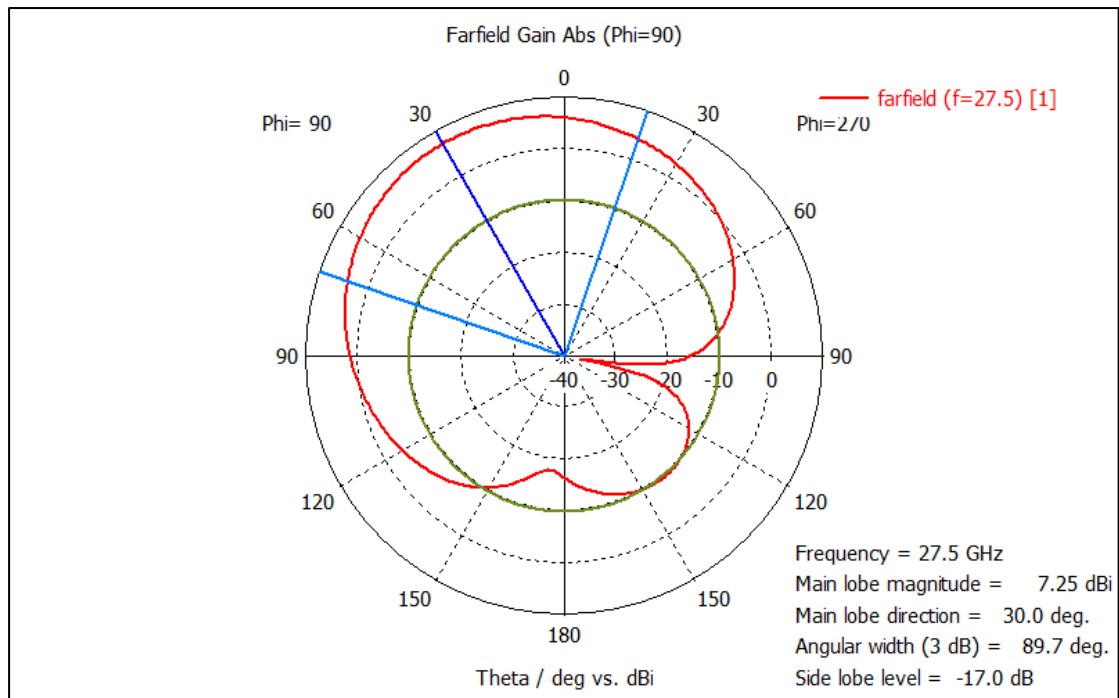


Figure 4.6 Gain polar plot

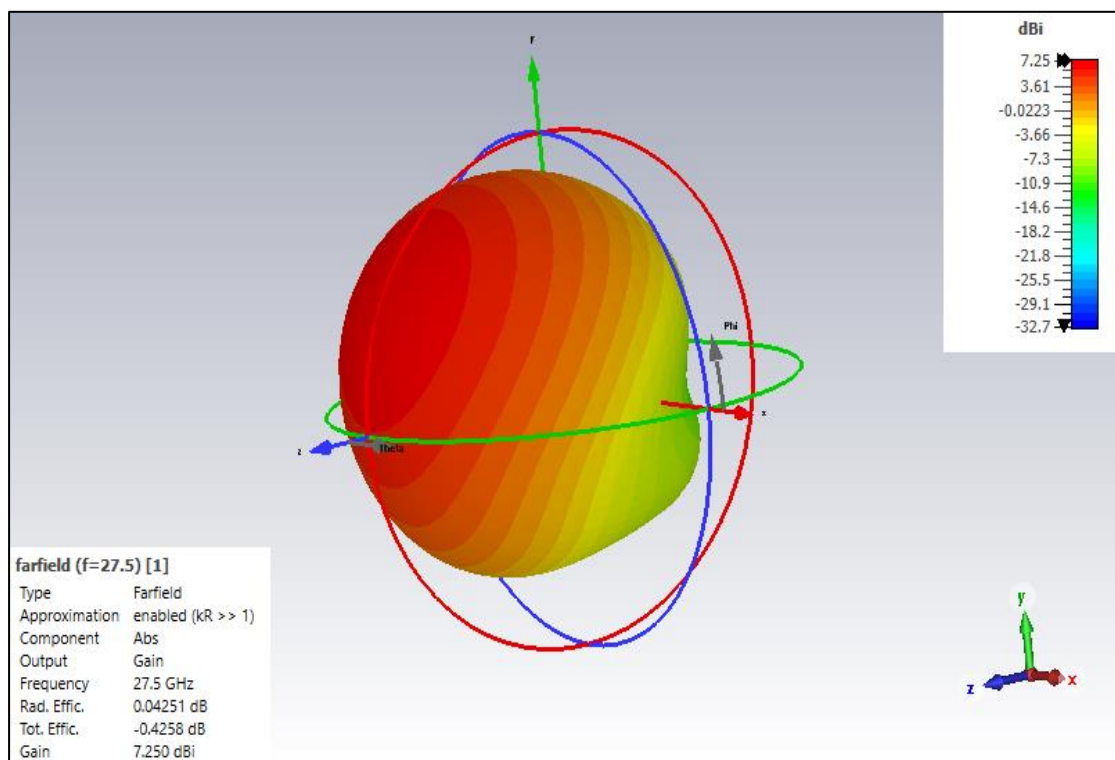


Figure 4.7 3D View of single element illustrating gain

Figure 4.7 illustrates radiation pattern of single element antenna and its gain, also we can observe directivity along with 3D radiation pattern of antenna in figure 4.8.

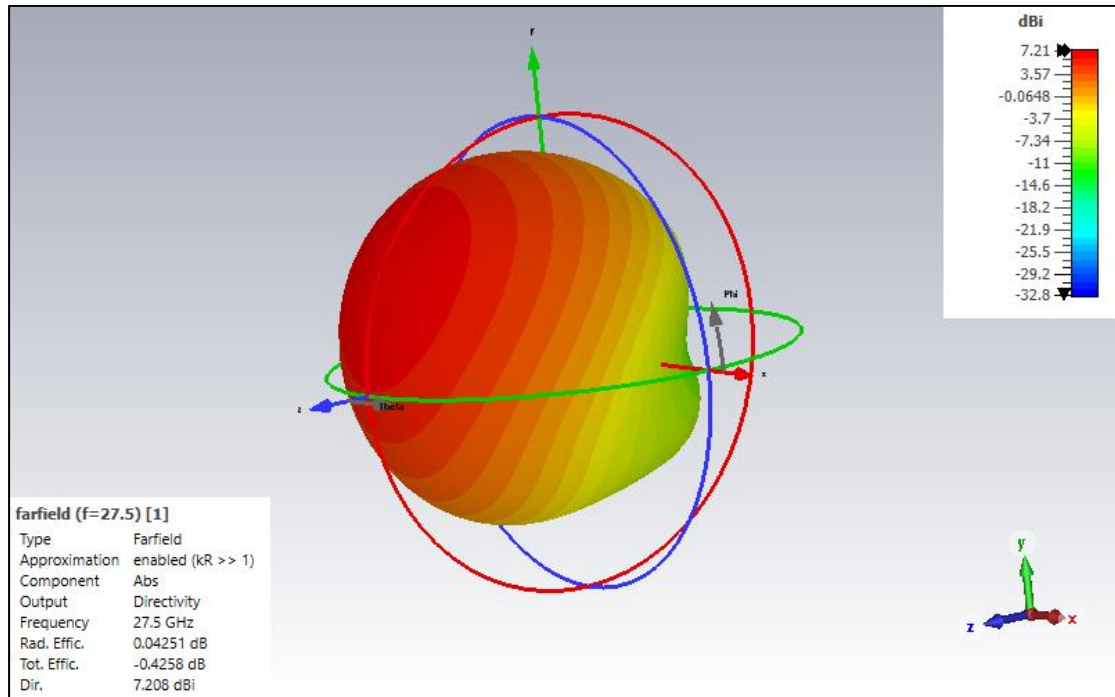


Figure 4.8 Directivity and 3D radiation pattern of single element

4.3 Transition from single element to 2- Element array

4.3.1 Design Dimensions and optimization

As we discussed in results of single element that it have a very low return loss, vswr and a fair gain and directivity values, but still there remains issues such as undesired resonant frequency, gain and directivity are not upto standard of 5G applications, so we made following changes in design as illustrated in figure 4.9 to design 2-element array. Results of this design are discussed afterwards. Figure 4.10 shows 2-Element Array with mentioned changes in dimensions as per transition from single element of antenna.

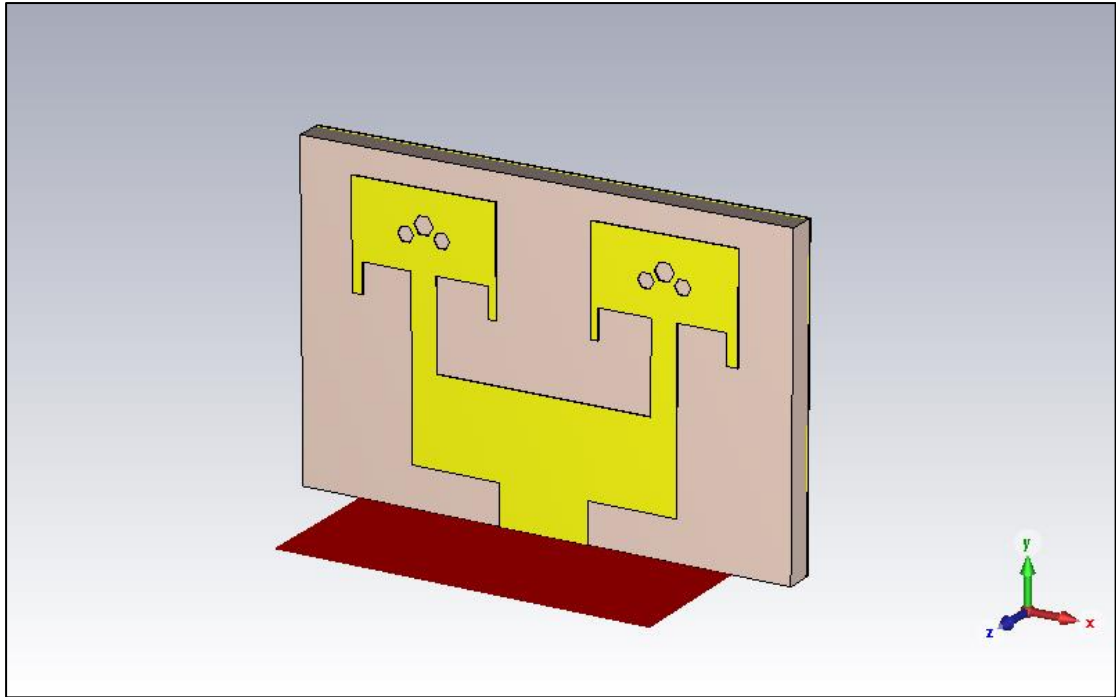


Figure 4.9 2-Element Array

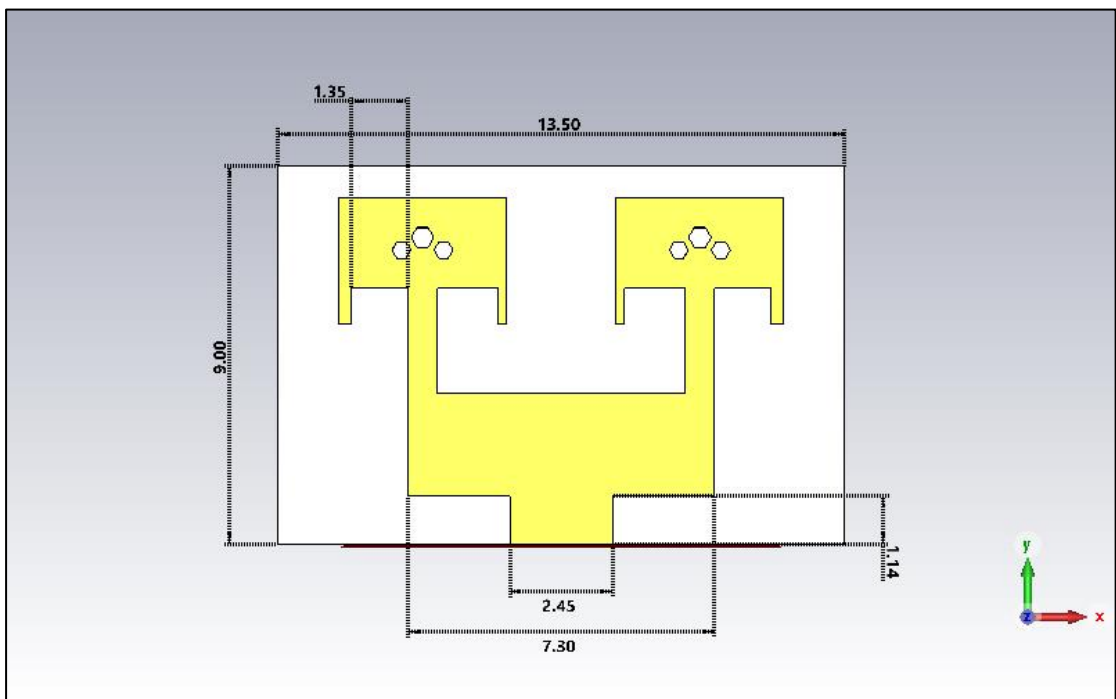


Figure 4.10 2-Element array with dimensions

Results of 2-Element array are as discussed in next section as followed.

4.3.2 Reflection co-efficient and resonant frequency

Transitioning from a single microstrip patch antenna element to a 2-element array introduces notable improvements in key performance metrics such as gain and directivity. The single patch antenna, resonating at 27.523 GHz with an excellent return loss of -32.66 dB, already demonstrates good impedance matching and efficient radiation characteristics. The S11 parameter as shown in figure 4.11 indicates minimal power reflection at this resonance frequency, confirming effective matching between the feed line and the patch. Upon configuring the antenna into a 2-element array, the collective radiation pattern becomes more focused due to the constructive interference of fields from each element, leading to an increase in both gain and directivity. Additionally, the impedance at the feed point remains well matched, measured close to 49.5 ohms, which ensures minimal mismatch losses and efficient power transfer from the transmission line to the array. This impedance consistency is crucial to maintain the antenna's broadband performance and overall efficiency as the array configuration evolves.

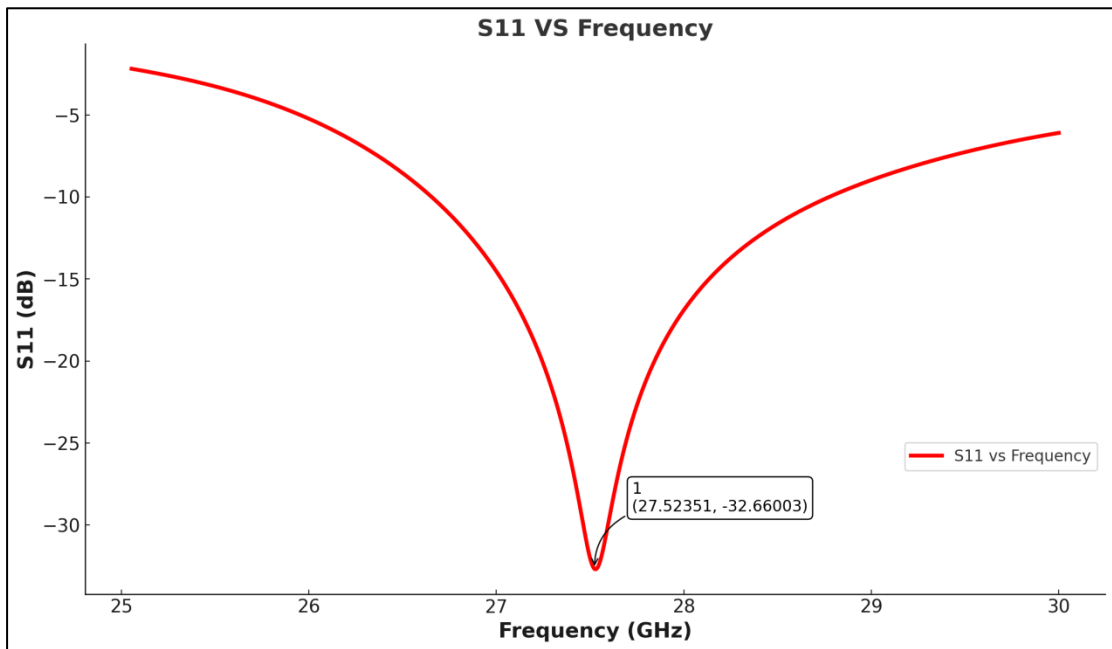


Figure 4.11 S11-Frequency graph of 2-Element Array

4.3.3 Bandwidth

The 2-element microstrip patch antenna array exhibits a bandwidth that extends from 26.65 GHz to 28.77 GHz as shown in figure 4.12, offering an operational frequency

range of approximately 2.12 GHz below -10db. This bandwidth is a significant improvement over the single-element antenna, reflecting the beneficial effects of array configuration on frequency response. The broader bandwidth ensures that the antenna can accommodate wider signal spectra, which is essential for high-data-rate applications such as 5G mmWave communications. Maintaining good impedance matching across this range allows the antenna array to perform efficiently without significant power loss or reflection, ensuring consistent signal quality. The expanded bandwidth, combined with enhanced gain and directivity from the array setup, provides a robust platform capable of supporting the demanding requirements of next-generation wireless systems.

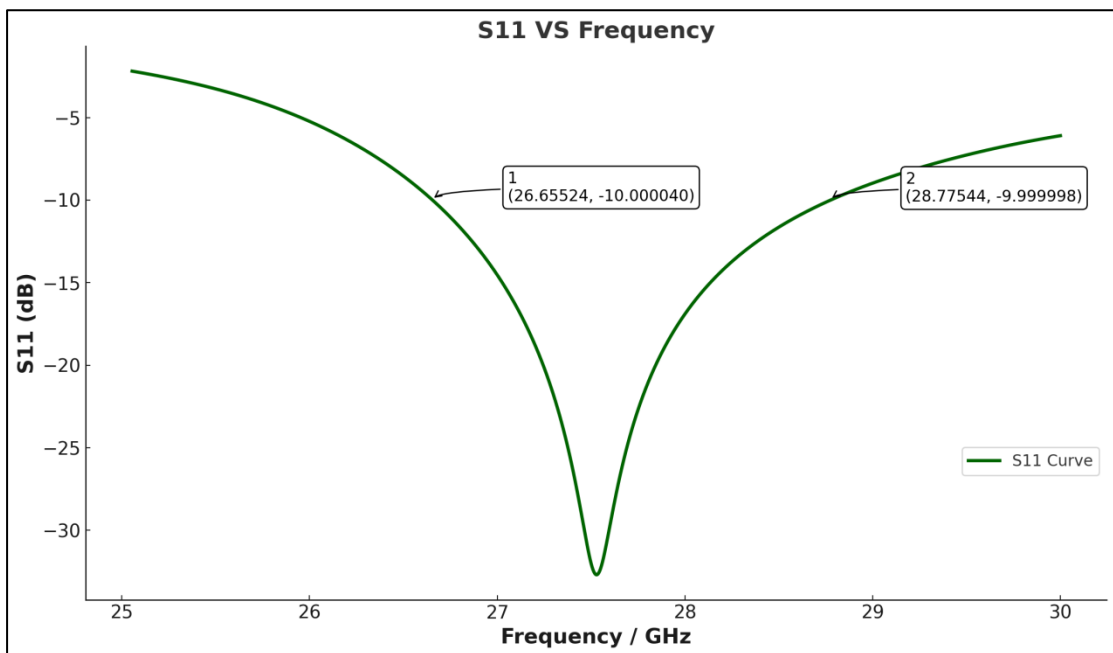


Figure 4.12 Bandwidth points marked on S11-freq graph

4.3.4 VSWR

At 27.528 GHz, the 2-element antenna array exhibits a Voltage Standing Wave Ratio (VSWR) of approximately 1.048, indicating an excellent impedance match between the antenna and the feeding transmission line.

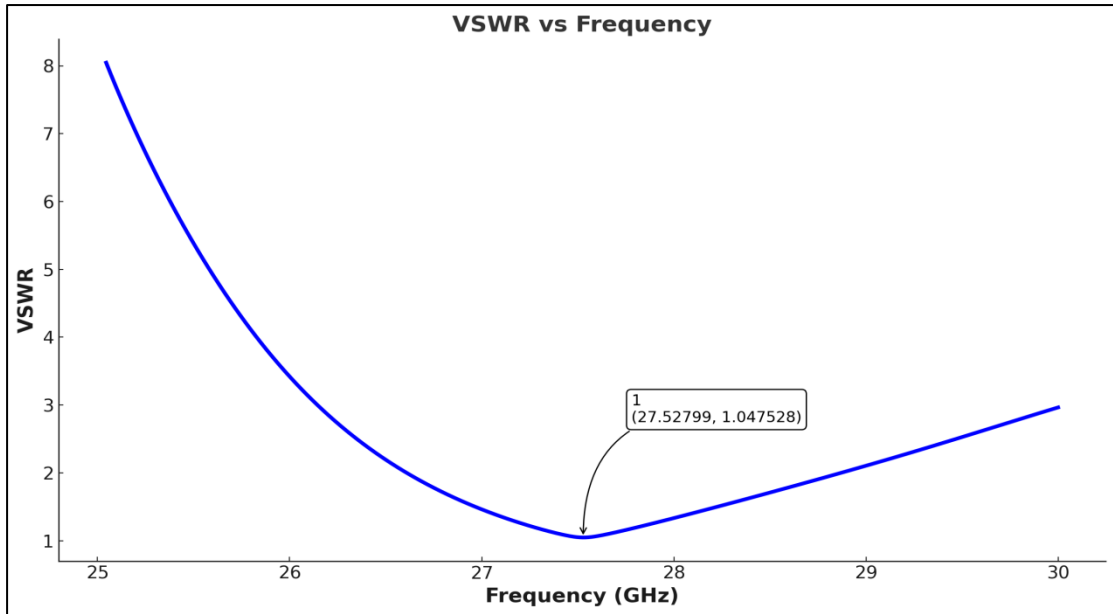


Figure 4.13 VSWR of 2-Element array

A VSWR value close to 1 signifies that nearly all the power supplied to the antenna is effectively radiated with minimal reflections. This ensures efficient power transfer and minimal losses at this key operating frequency, which is crucial for maintaining strong and reliable signal transmission in 5G mmWave systems. The VSWR over the frequency band, highlighting this optimal matching at 27.528 GHz, is illustrated in Figure 4.13

4.3.5 Efficiency

The efficiency of the 2-element antenna array is calculated by dividing the radiated power by the input power. With a radiated power of 0.4226 watts and an input power of 0.5 watts as illustrated and marked on curves in Figure 4.14, the efficiency is 84.52%.

$$\eta = \frac{P_{\text{radiated}}}{P_{\text{input}}} = 0.4226/0.5$$

$$\eta = 0.845 = 84.5\%$$

This efficiency indicates that approximately 84.5% of the input power is effectively converted into radiated electromagnetic energy, while the remaining power is lost due to dielectric, conductor, or other losses. Although slightly lower than the single-element antenna efficiency, this level remains quite good, especially considering the increased complexity of the array configuration. Such efficiency is suitable for mmWave 5G applications where a balance between gain, bandwidth, and power utilization is critical.

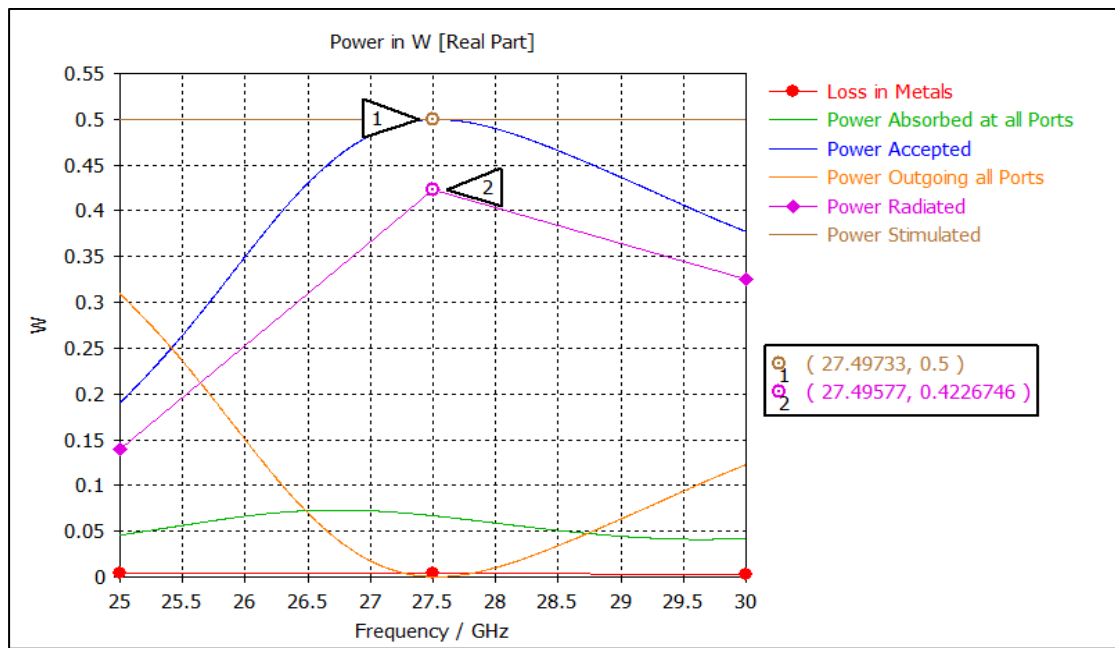


Figure 4.14 Showing curves of different powers and losses of 2-Element

4.3.6 Gain and Directivity

The 2-element antenna array exhibits a directivity of 8.24 dBi and a gain of 7.52 dBi, reflecting its ability to effectively concentrate radiated power in the desired direction while accounting for some practical efficiency losses. The directivity and gain patterns are clearly illustrated in the polar plots shown in Figure 4.15 and Figure 17-4.16, respectively, which provide insight into the angular distribution of radiated energy. Additionally, the 3D radiation patterns depicted in Figure 4.17 for gain and Figure 4.18 for directivity further visualize the spatial radiation characteristics, highlighting the antenna's focused beam and confirming the measured performance values.

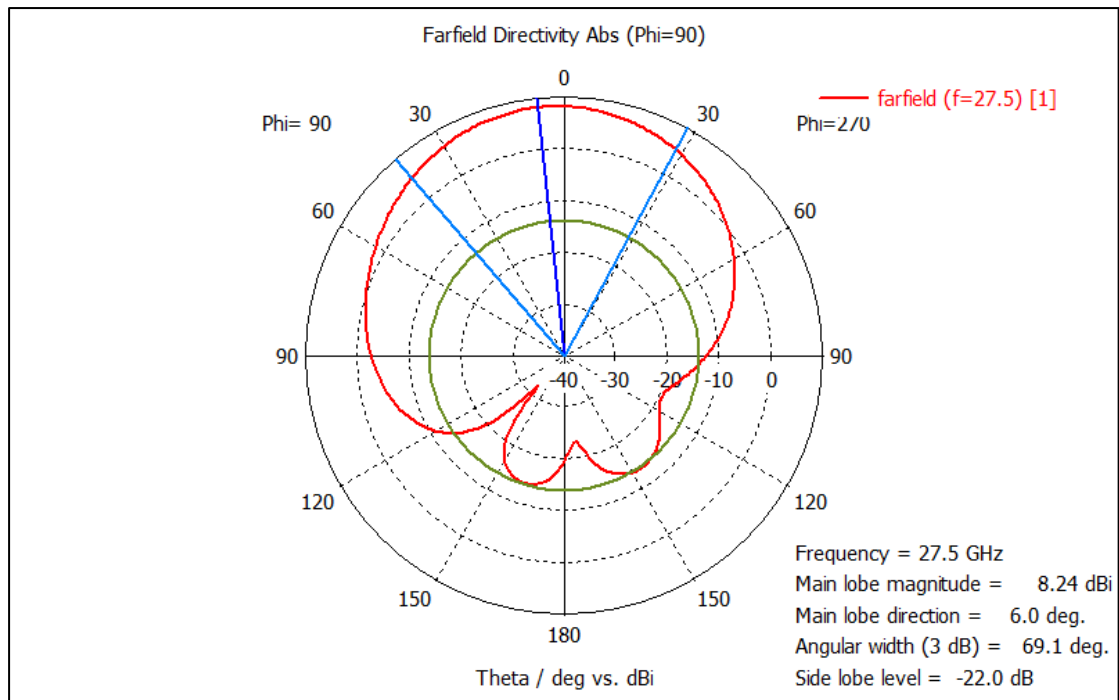


Figure 4.15 Directivity of 2 element shown in polar plot

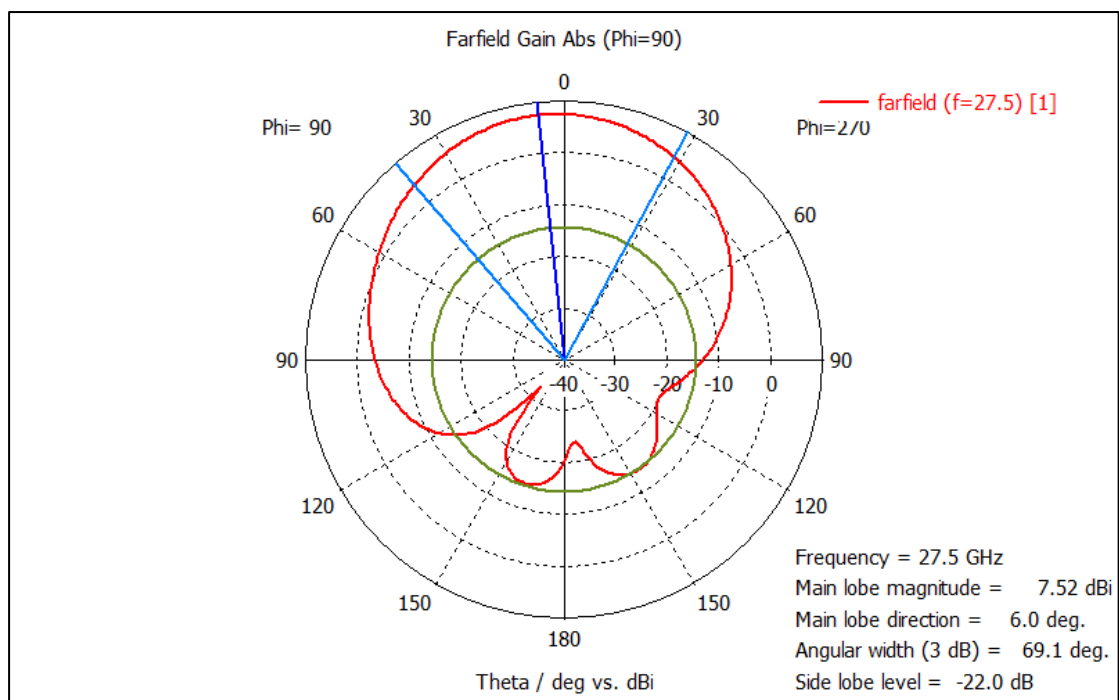


Figure 4.16 Gain of 2 element shown in polar plot

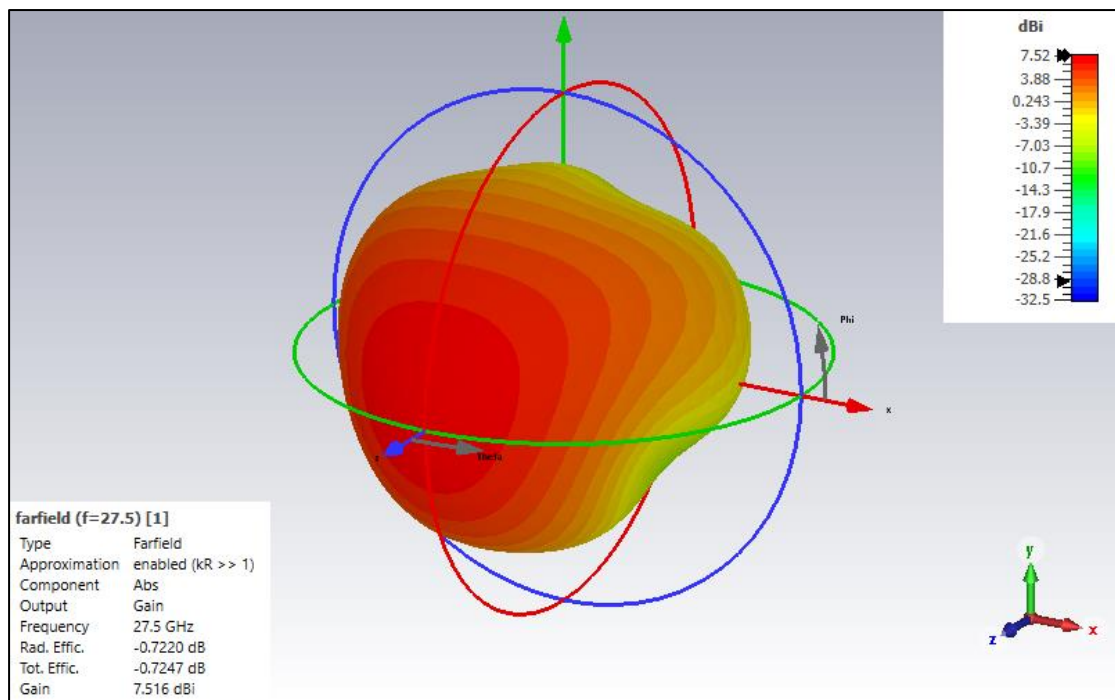


Figure 4.17 3D radiation pattern of 2-element antenna with gain values

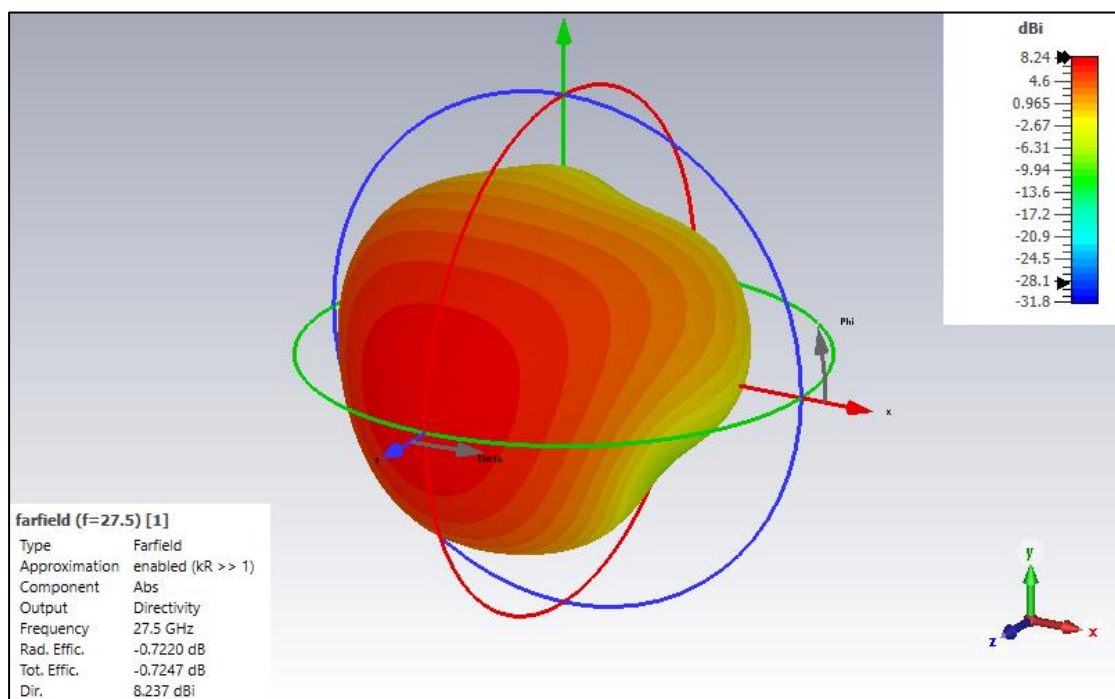


Figure 4.18 3D radiation pattern of 2-element with directivity values

4.4 Design and Analysis of 4-Port MIMO Antenna

4.4.1 Motivation and Design Objectives

The reason for extending the antenna design from a 2-element array to a 4-port MIMO configuration arises from the growing demand for higher data rates and improved link reliability in 5G communications due to low ECC, high gain, high directivity and low losses. A 4-port MIMO antenna enables simultaneous transmission and reception of multiple independent data streams, significantly enhancing channel capacity and spectral efficiency. This configuration also improves spatial diversity, reducing signal fading and increasing communication robustness in complex propagation environments.

The primary objectives of the 4-port antenna design are to achieve compactness suitable for practical 5G device integration, while maintaining high gain, wide bandwidth, and excellent isolation between antenna elements. Ensuring good impedance matching across all ports and minimizing mutual coupling are critical to optimizing overall system performance. The design also aims to maintain consistent radiation patterns and low envelope correlation coefficients to fully leverage the benefits of MIMO technology.

4.4.2 Design Dimensions

The 4-port MIMO antenna design is based on duplicating and rotating the previously designed 2-element antenna configuration. The overall ground plane and substrate have dimensions of 35.58 mm by 35.58 mm, providing a compact platform for the array. Each of the original 2-element antenna dimensions is preserved, with the two-element unit copied and rotated appropriately to form the 4-port structure. The duplicated units are positioned such that each is placed 6 mm away from the center of each side of the substrate, ensuring symmetrical element placement. The distance between two consecutive antenna elements is maintained at 10.1 mm to balance mutual coupling and compactness, while the horizontal spacing D between elements is set at 19 mm as shown in figure 4.21. This arrangement aims to optimize isolation and overall antenna performance within the constrained substrate size, facilitating efficient operation in the 28 GHz mmWave band.

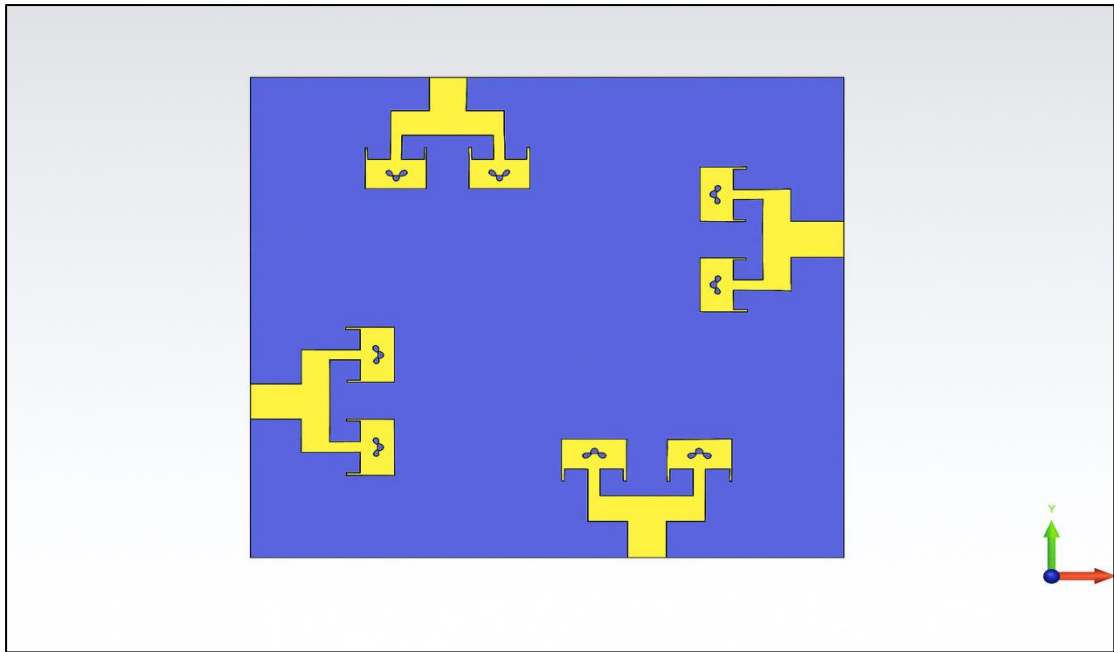


Figure 4.19 2D view of 4-port MIMO antenna

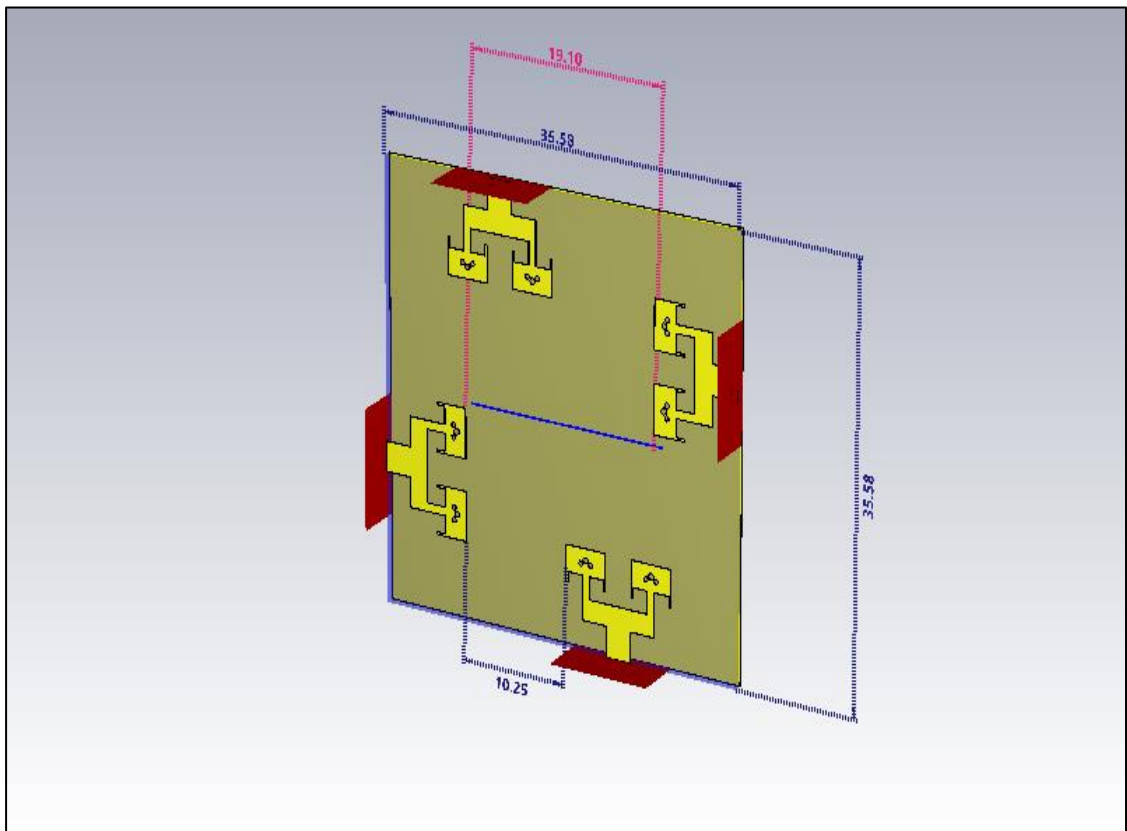


Figure 4.20 3D MIMO view with dimensions

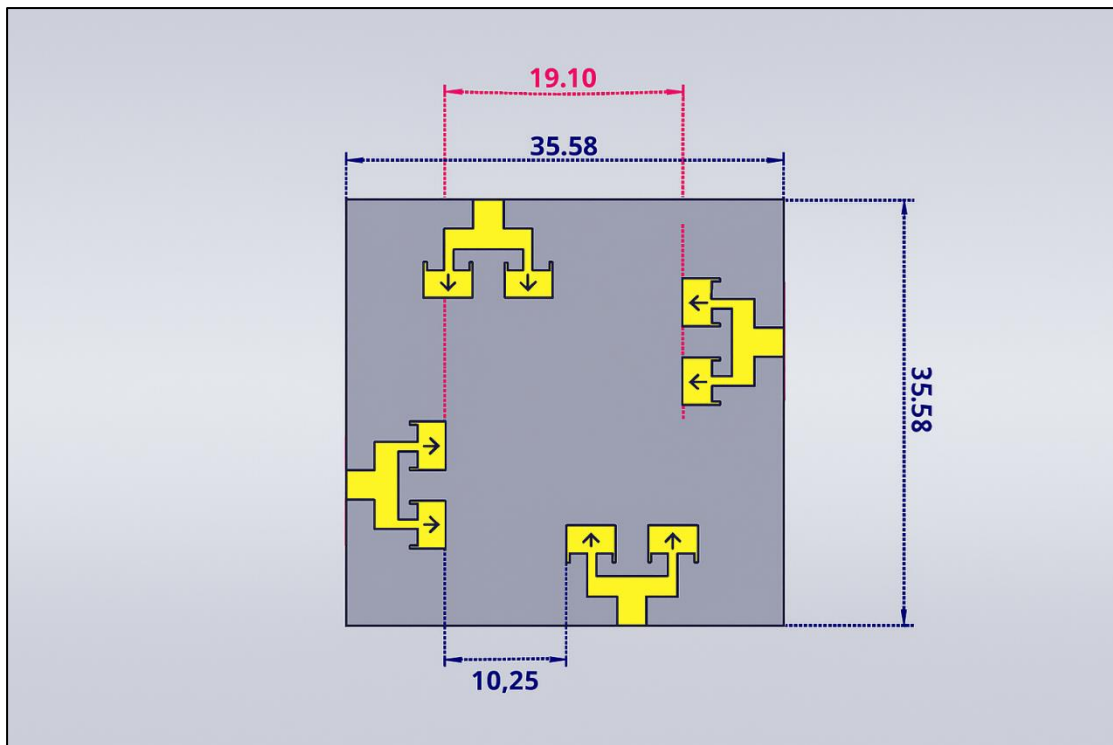


Figure 4.21 2D MIMO view with dimensions

4.4.3 Reflection coefficient and resonant frequency of MIMO

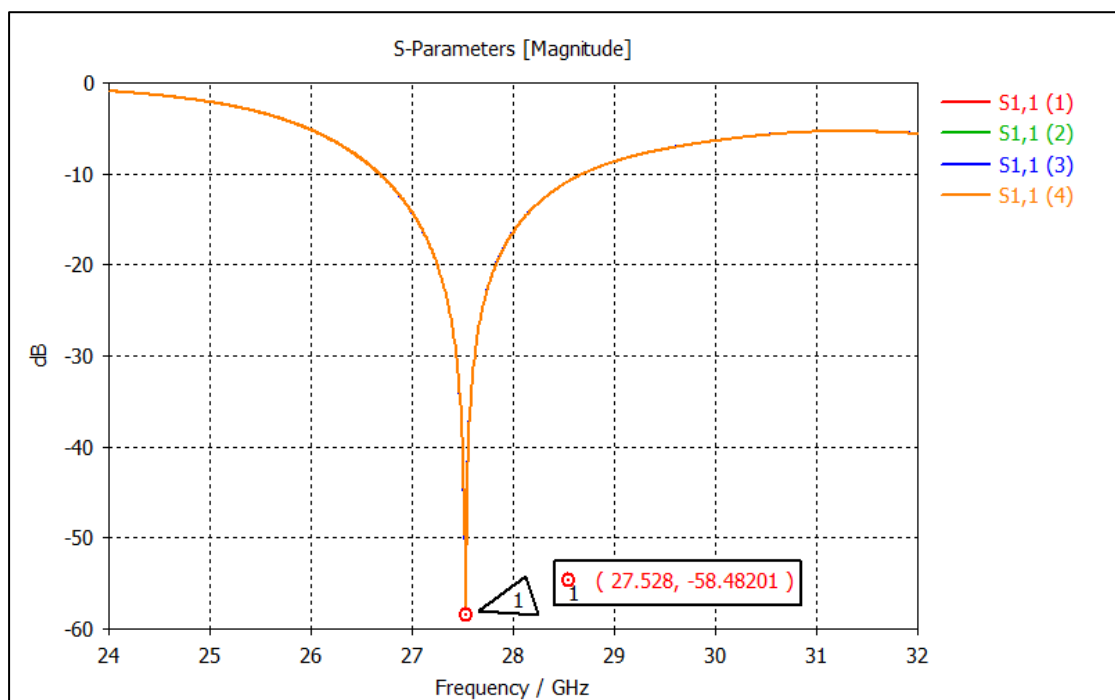


Figure 4.22 S11 and frequency of MIMO

The reflection coefficient S_{11} of the 4-port MIMO antenna shows a sharp resonance at 27.528 GHz, indicating the frequency at which the antenna is best matched to the feeding transmission line. At this resonance frequency, the antenna exhibits an impressive return loss of approximately -58.48 dB, signifying that almost all the input power is effectively delivered to the antenna with minimal reflection. Also all the reflection coefficient graphs are almost equal curves as $S_{11}=S_{22}=S_{33}=S_{44}$ of all ports. Such a low return loss demonstrates excellent impedance matching, which is critical for efficient radiation and reducing power wastage. The S_{11} performance across the frequency band, highlighting this resonance and return loss, is illustrated in Figure 4.22, clearly showing the antenna's strong capability to operate efficiently at the designated 5G mmWave frequency.

4.4.4 Bandwidth

The bandwidth of the 4-port MIMO antenna is defined by the frequency range over which the reflection coefficient S_{11} remains below the acceptable threshold, typically -10 dB. In this design, the bandwidth of MIMO is 2.001 GHz antenna exhibits a bandwidth extending from approximately 26.676 GHz to 28.687 GHz, covering a wide frequency span suitable for 5G mmWave applications. This bandwidth range ensures that the antenna can efficiently transmit and receive signals across multiple channels within the 28 GHz band, supporting high data rates and reliable connectivity. The exact bandwidth points are clearly marked on the S_{11} graph, as illustrated in Figure 24-4.4.4, which visually emphasizes the antenna's effective operational frequency range.

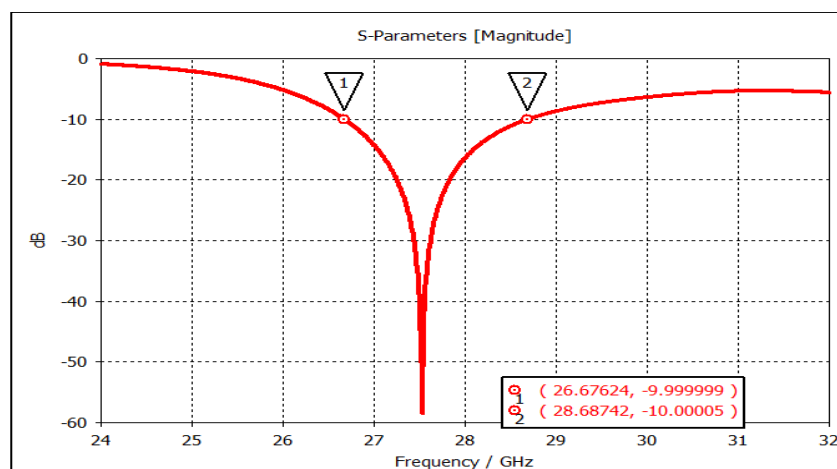


Figure 4.23 illustrating bandwidth points marked in s_{11} graph

4.4.5 Transmission Co-efficients and Isolation

The transmission coefficients S_{ij} between the ports of the 4-port MIMO antenna represent the amount of power transmitted from port i to port j , which is crucial for assessing the mutual coupling and isolation between antenna elements. For an effective MIMO system, these transmission coefficients should be as low as possible, indicating minimal coupling and interference between ports. In our design, all transmission coefficients—including $S_{12}, S_{21}, S_{13}, S_{31}, S_{14}, S_{41}$ etc are consistently below -25 dB across the operating bandwidth. This excellent isolation ensures that antenna elements operate independently with very low mutual interference, which significantly improves signal quality and capacity for simultaneous multi-stream transmission. The transmission coefficient results are clearly illustrated in the S-parameter plots, confirming the strong isolation performance of the antenna system.

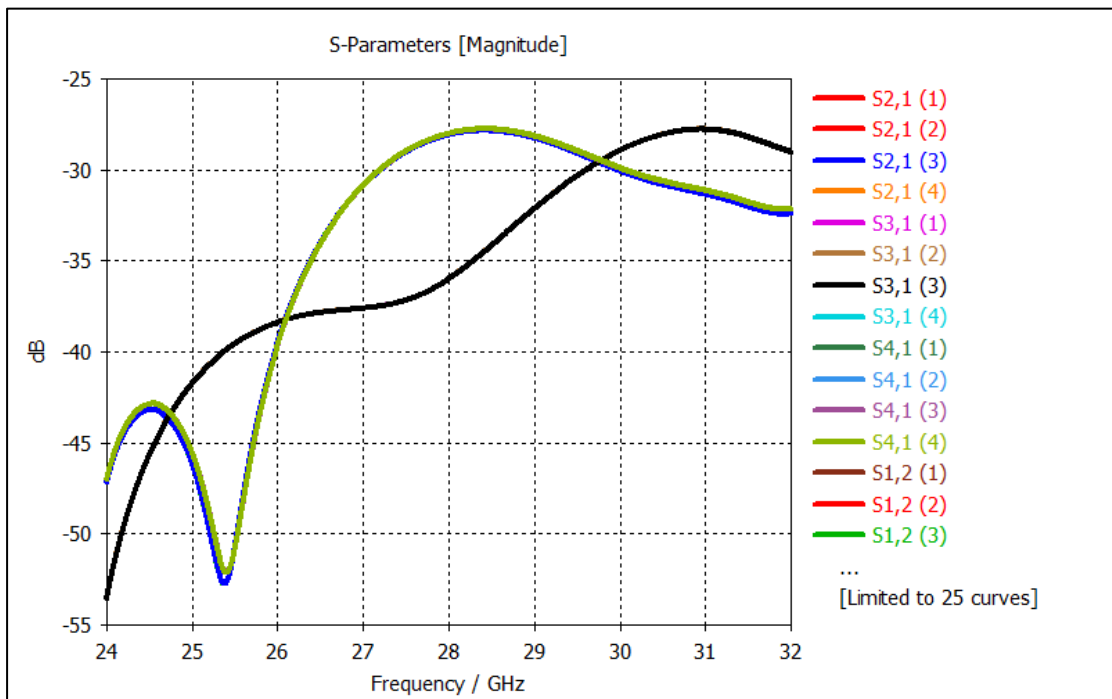


Figure 4.24 Showing graph of transmission co-efficients

4.4.6 VSWR of MIMO

The Voltage Standing Wave Ratio (VSWR) of all ports in the 4-port MIMO antenna is exceptionally low, with values around 1.0025, indicating excellent impedance matching between each antenna port and the feeding transmission line. The

impedance at the feed point is closely matched to 49.5 ohms, which is near the standard 50-ohm line impedance, ensuring efficient power transfer with minimal reflections. This effective impedance matching reduces signal loss and enhances overall antenna performance. The impedance matching behavior, along with the VSWR characteristics, is clearly illustrated in Figure 2-4.4.6 and Figure 27.4.4.6, demonstrating the antenna’s suitability for 5G mmWave applications where maintaining signal integrity is critical.

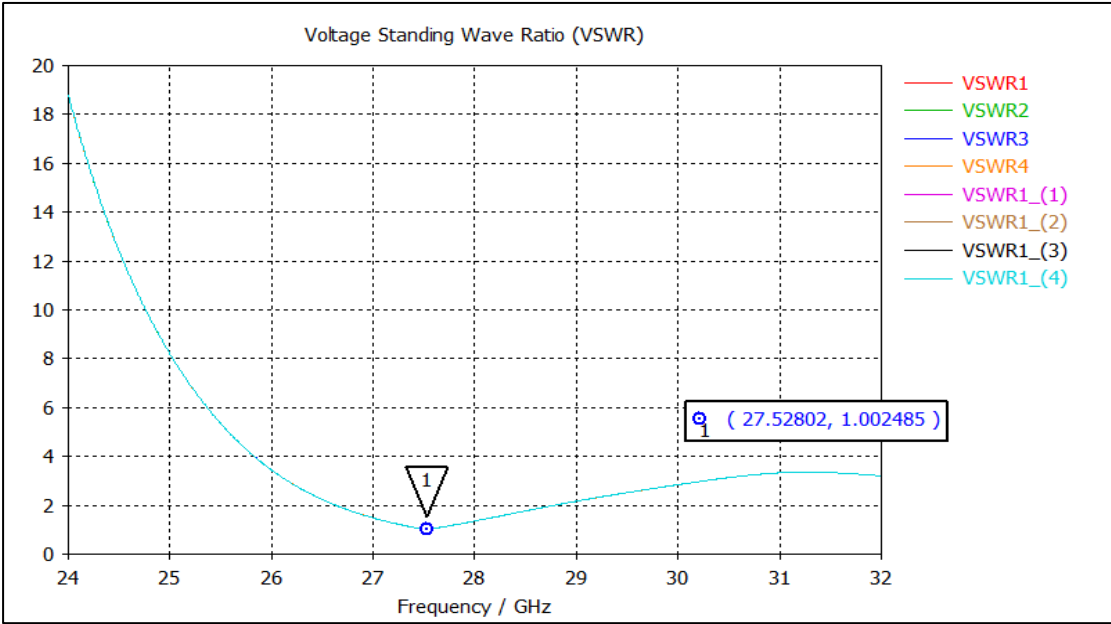


Figure 4.25 VSWR of MIMO

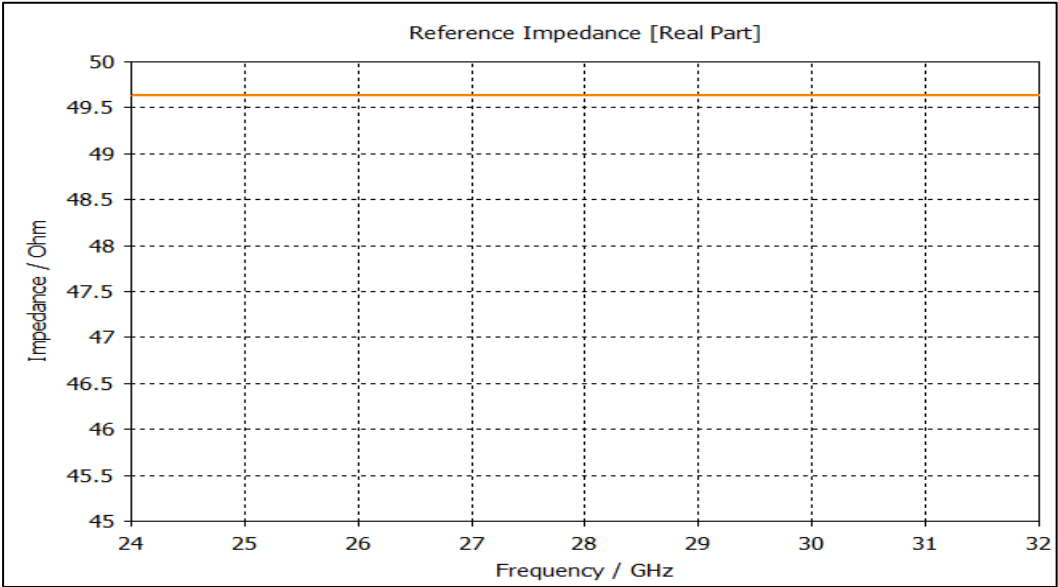


Figure 4.26 impedance graph

4.4.7 Gain And Directivity of MIMO

The gain and directivity of the 4-port MIMO antenna are critical indicators of its ability to focus radiated energy and enhance signal strength in desired directions. In this design, each antenna element within the MIMO array exhibits a gain of 8.32 dBi and a directivity of 9.13 dBi, indicating high radiation efficiency and minimal losses. The combined effect of multiple elements in the MIMO configuration results in improved overall gain compared to a single element, enabling stronger and more reliable signal transmission and reception.

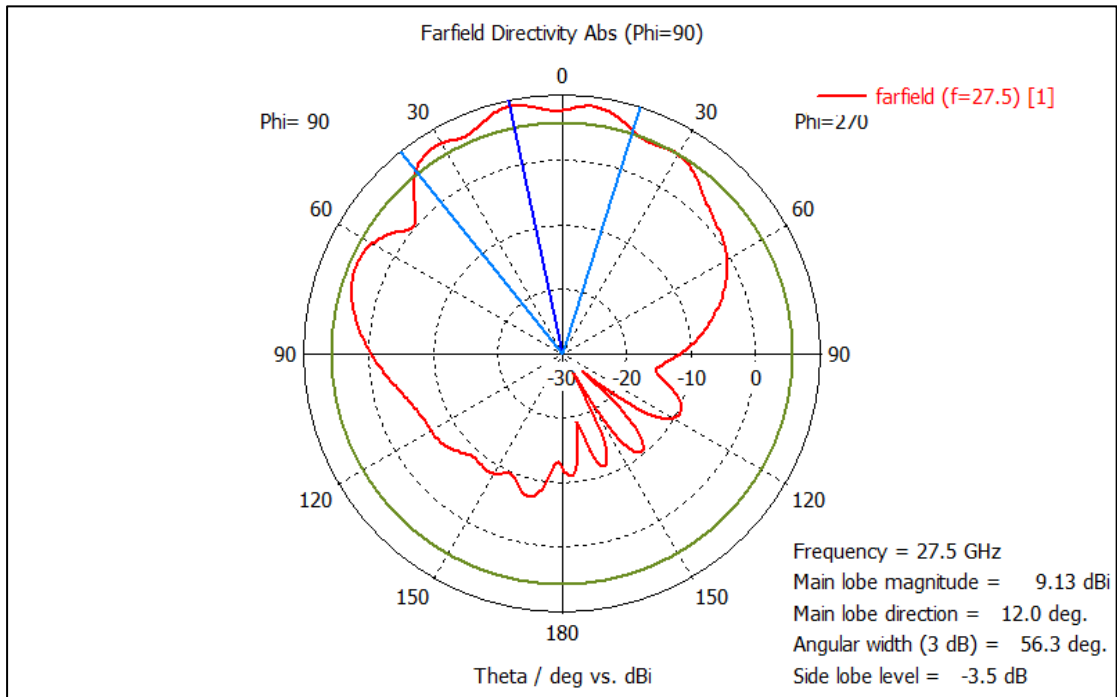


Figure 4.27 polar plot for directivity

High directivity ensures that the antenna concentrates electromagnetic energy effectively, which is essential at mmWave frequencies to overcome increased path loss. The gain and directivity characteristics are clearly illustrated in the polar plots shown in Figure 29-4.4.7 for directivity and Figure 30-4.4.7 for gain. Additionally, the 3D radiation patterns depicted in Figure 31-4.4.7 for gain and Figure 32-4.4.7 for directivity provide comprehensive visualization of the antenna's focused main lobes and low side lobes, contributing to enhanced performance in complex wireless environments.

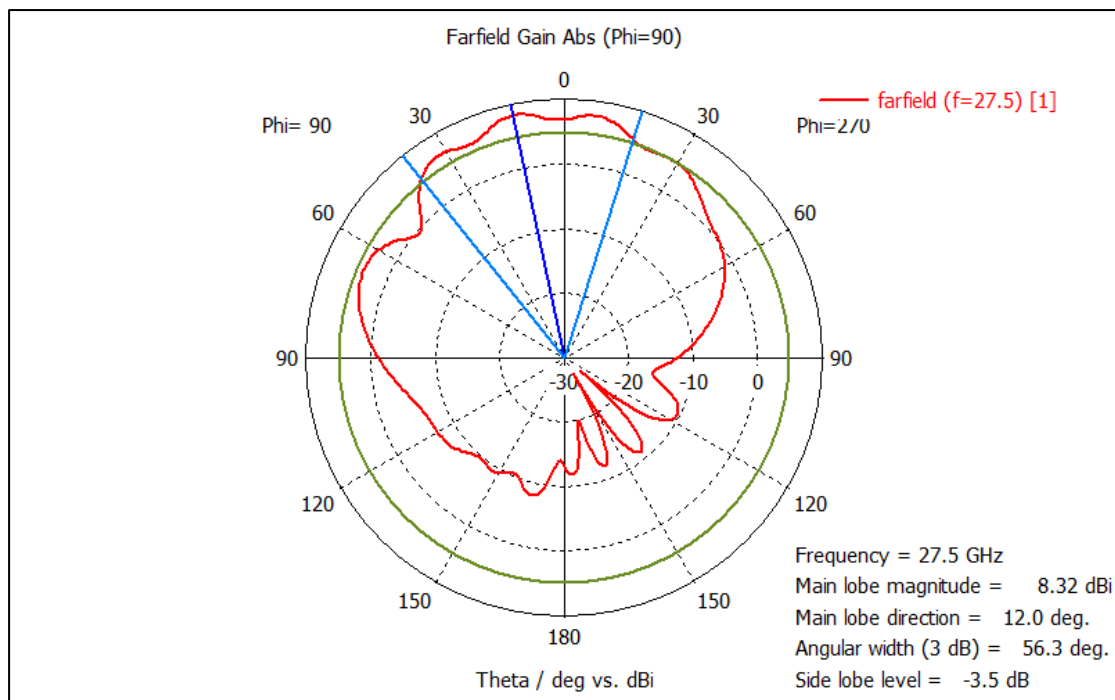


Figure 4.28 Polar Plot of Gain

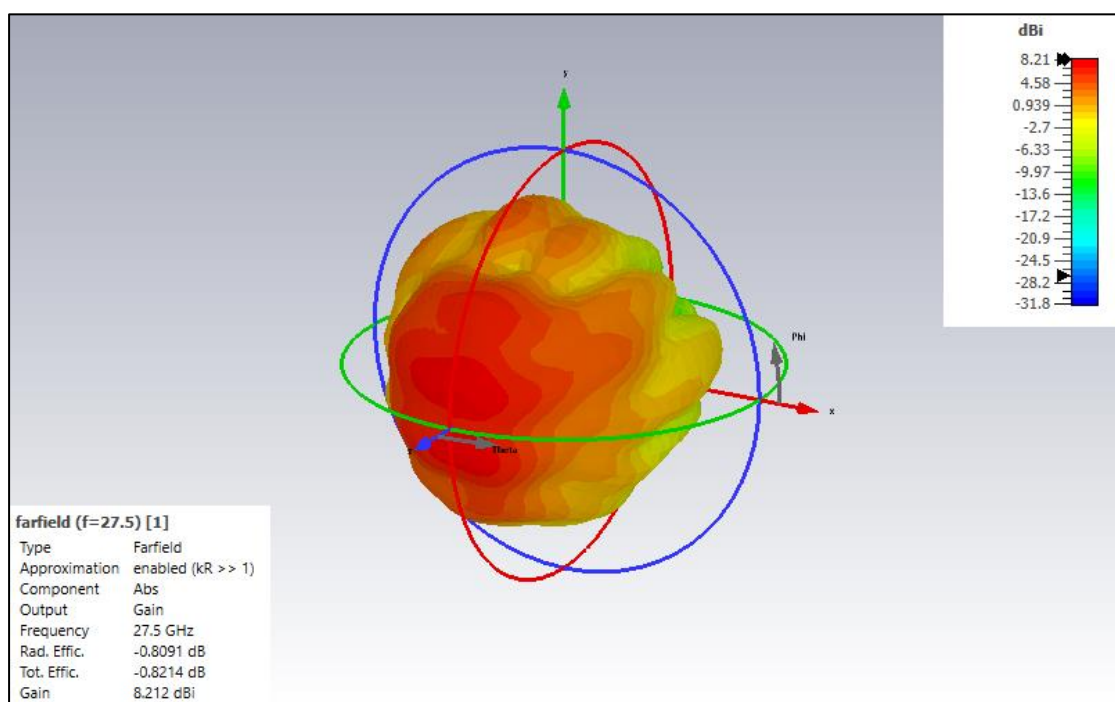


Figure 4.29 3D view of Gain

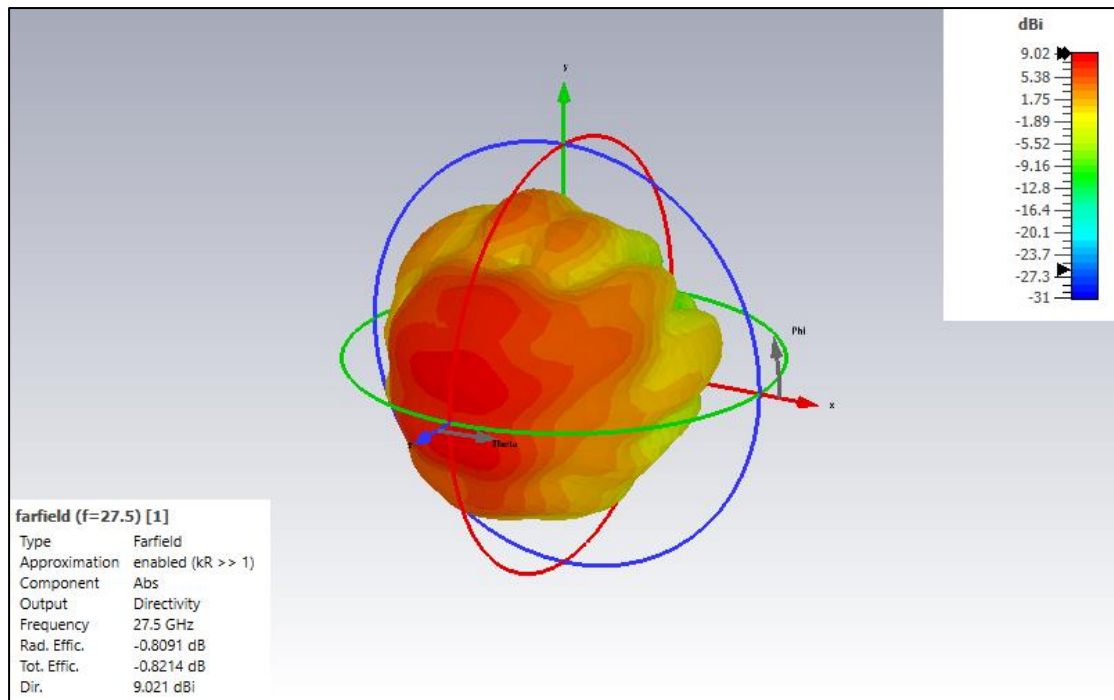


Figure 4.30 3D view of directivity

4.4.8 Envelope Correlation Coefficient (ECC)

At the resonant frequency of 27.528 GHz, the envelope correlation coefficient (ECC) of the MIMO antenna is approximately 8.68×10^{-9} , which is an exceptionally low value. This near-zero ECC indicates that the antenna elements operate with almost complete independence, providing highly uncorrelated signals. Such a low ECC reflects excellent spatial diversity, ensuring that the MIMO system can efficiently support multiple simultaneous data streams with minimal interference. This level of performance is critical in 5G applications. The graph of ECC, showing the value at this frequency and across the frequency range, is illustrated in Figure 33-4.4.8, providing a clear visualization of the antenna's performance.

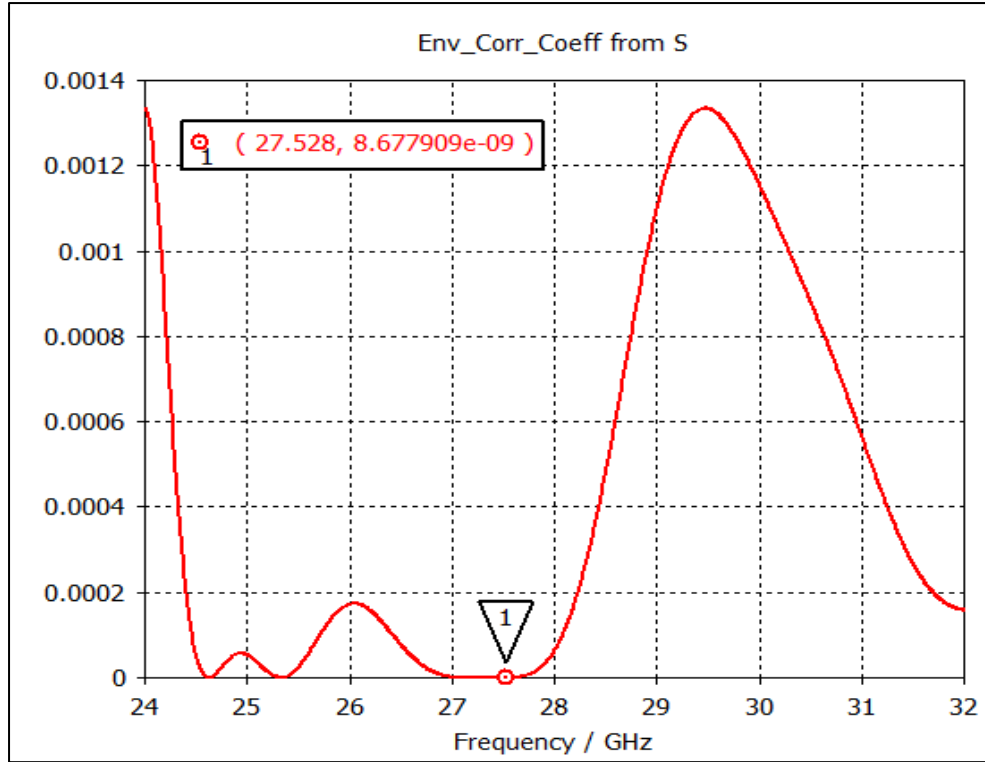


Figure 4.31 graph of ECC

4.4.9 Diversity Gain (DG)

Diversity Gain (DG) is a measure of the improvement in signal quality due to spatial diversity in a MIMO system. It is calculated from the Envelope Correlation Coefficient (ECC) using the following formula:

$$DG = 10 \times \sqrt{1 - (ECC)^2}$$

ECC is the Envelope Correlation Coefficient.

Calculation of Diversity Gain

Given that the ECC value at the resonant frequency of 27.528 GHz is 8.68×10^{-9} , we can calculate the Diversity Gain as follows:

$$DG = 10 \times \sqrt{1 - (8.68 \times 10^{-9})^2}$$

$$DG = 10$$

The calculated Diversity Gain (DG) of 10 dB reflects an optimal MIMO system performance with very low signal correlation between antenna elements. This indicates that the antenna elements are well-spaced in terms of received signal diversity, effectively improving the overall system's ability to mitigate fading and multipath effects. Such high diversity gain is essential for ensuring robust

communication in challenging mmWave environments, particularly in 5G systems, where high data rates and reliability are crucial.

Table 4.1 Comparison of Single Element ,2-Element and MIMO

Parameter	Single Element	2-Element Array	4-Port MIMO
Resonant Frequency (GHz)	29.1	27.523	27.528
Return Loss (dB)	-64	-32.66	-58.48
VSWR	1.00113	1.048	1.0025
Bandwidth (GHz)	2	2.12	2.001
Gain (dBi)	7.25	7.52	8.32
Directivity (dBi)	7.21	8.24	9.13
Efficiency (%)	94.6	84.5	86
ECC	N/A	N/A	8.68×10^{-9}

4.10 Dataset Collection

After completing the detailed antenna designs and simulations for the single element, 2-element array, and the final 4-port MIMO antenna configurations, the next critical phase involved building a comprehensive dataset for machine learning model training. For this, the finalized 4-port MIMO antenna design was simulated approximately 320 times in CST Microwave Studio. Each simulation iteration involved varying key input design parameters such as the width and length of the substrate (W_s and L_s), the width of the patch (W_p), and the width and length of the inset feed (W_i and L_i). For every variation, important output parameters were recorded, including return loss (S_{11}), resonant frequency (f_r), gain (G), directivity (D), and voltage standing wave ratio (VSWR). This iterative dataset forms a rich collection of input-output mappings essential for training machine learning regression models aimed at accurately

predicting antenna performance metrics based on design variables. By automating this parametric sweep and collecting simulation results systematically, the dataset serves as a robust foundation to explore how machine learning can accelerate and optimize antenna design, reducing reliance on time-consuming manual tuning and simulations. Dataset is attached as Table 4.2

Table 4.2 dataset Used For ML Model Training

#	A	B	C	D	E	F	G	H	I	J	K
1	Ws	ls	wp	wi	li	s11	fr	G	D	VSWR	
2	30	30	4	1.45	0.84	-33.70798406	27.50014	7.855	8.1	1.0242	
3	30.031	30.031	4	1.45	0.84	-33.98586924	27.48252	7.881	8.098	1.025	
4	30.063	30.063	4	1.45	0.84	-35.08680934	27.46991	7.874	8.097	1.0257	
5	30.094	30.094	4	1.45	0.84	-35.58237164	27.52289	7.908	8.097	1.0263	
6	30.125	30.125	4	1.45	0.84	-35.80304931	27.50498	7.988	8.092	1.027	
7	30.157	30.157	4	1.45	0.84	-35.71983376	27.48651	7.976	8.091	1.0276	
8	30.188	30.188	4	1.45	0.84	-36.70514551	27.47305	7.964	8.09	1.0285	
9	30.219	30.219	4	1.45	0.84	-36.97117378	27.52856	8.047	8.098	1.0291	
10	30.251	30.251	4	1.45	0.84	-36.97760005	27.51017	8.087	8.096	1.0299	
11	30.282	30.282	4	1.45	0.84	-36.48343664	27.49601	8.062	8.094	1.0307	
12	30.313	30.313	4	1.45	0.84	-36.66360368	27.4841	8.091	8.097	1.0313	
13	30.345	30.345	4	1.45	0.84	-37.82451731	27.52809	8.066	8.094	1.0319	
14	30.376	30.376	4	1.45	0.84	-36.49643599	27.51654	8.042	8.092	1.0327	
15	30.408	30.408	4	1.45	0.84	-37.45287987	27.50324	8.055	8.093	1.0334	
16	30.439	30.439	4	1.45	0.84	-36.61060761	27.49218	7.954	8.084	1.0339	
17	30.47	30.47	4	1.45	0.84	-37.65012193	27.50895543	7.863	8.075	1.0341	
18	30.502	30.502	4	1.45	0.84	-37.53615996	27.51008411	7.834	8.072	1.0335	
19	30.533	30.533	4	1.45	0.84	-36.39059982	27.51121279	7.78	8.067	1.0331	

4.5 Machine Learning Model Training

In addition to full-wave simulation, the gain and directivity values were also predicted using machine learning models developed earlier in the project. The Random Forest Regressor, identified as the most accurate model, predicted gain and directivity with a mean absolute error of 2.32% and an R^2 score of 99.07%, closely matching the CST results. This validates the reliability of the surrogate model for future antenna optimization tasks.

4.5.1 Model Training and Evaluation for Predictive Modeling

This chapter details the process of developing and assessing a predictive model designed to estimate the target variable 'D' using selected input features. The methodology follows a standard machine learning pipeline, from initial data handling to model evaluation.

4.5.2 Data Loading and Initial Assessment

The process commenced with loading the raw dataset from an Excel file located at /content/dataset.xlsx. This data was imported into a pandas DataFrame. To understand the dataset's characteristics, an analysis function was used, providing several key insights:

- **Dataset Structure:** The DataFrame contained 320 entries (rows) and 10 columns. All columns were of the float64 data type.
- **Data Completeness:** The analysis revealed that all columns had 320 non-null counts, indicating no missing values present in the dataset upon loading.
- **Summary Statistics:** Descriptive statistics for the numerical columns were generated. This included count, mean, standard deviation, minimum, maximum, and quartile values for variables like 'Ws', 'ls', 'wp', 'wi', 'li', 'S11', 'fr', 'G', 'D', and 'VSWR'. The mean value for the target variable 'D' was **8.101140**. The range of 'D' values observed was from a minimum of **7.830000** to a maximum of **9.710000**.
- **Missing Values Check:** An explicit check confirmed the absence of missing values, showing "Series([dtype: int64)]".
- **Duplicate Rows Check:** A count of duplicate rows was performed, identifying 0 duplicate rows.
- **Sample Data:** The first five rows were displayed, showing example entries for all 10 columns.

This initial assessment confirmed that the data was fully populated and free from duplicates, providing a sound basis for subsequent processing.

4.5.3 Data Cleaning: Outlier Detection and Removal

Although the dataset lacked missing values and duplicates, outliers were identified and removed to improve model robustness. The **Interquartile Range (IQR) method** was applied. This involved calculating Q1 and Q3 for each numerical column, determining the IQR, and defining bounds ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$). Rows containing values outside these bounds in any column were removed. This cleaning step resulted in a filtered DataFrame, df_cleaned, with **277 rows** remaining.

4.5.4 Feature Selection

From the `df_cleaned` dataset, the input features and the target variable were defined. The **target variable** for prediction was directivity of antenna '**D**'. The selected features used for training were '**Ws**', '**wp**', '**wi**', '**li**', '**VSWR**', and '**G**'. Columns '**S11**' and '**fr**' were explicitly excluded from the feature set.

4.5.5 Model Selection

For the regression task (predicting continuous variable '**D**'), a **Gradient Boosting Regressor** model from `sklearn ensemble` was chosen. This is an ensemble technique that builds predictive models sequentially, where each new model corrects errors made by prior models, generally leading to high accuracy.

4.5.6 Data Splitting

To evaluate the model's performance on unseen data, the cleaned dataset was split into training and testing sets. Using `train_test_split` from `sklearn model selection`, **30%** of the data was reserved for testing (`test_size=0.3`), and the remaining **70%** for training. A `random_state` of 42 was set to ensure the split is consistent across multiple runs, facilitating reproducibility. This produced `X_train`, `X_test`, `y_train`, and `y_test`.

4.5.7 Model Training

The Gradient Boosting Regressor model (`gb_model`) was initialized with `n_estimators=100` and `random_state=42`. The model was trained using the `X_train` and `y_train` data via the `fit()` method. This process involved the algorithm learning the relationship between the selected features and the target variable based solely on the training data.

4.5.8 Model Evaluation

After training, the `gb_model`'s performance was assessed using the test set (`X_test`, `y_test`). Predictions (`y_pred_gb`) were generated for `X_test`. Standard regression metrics were then calculated to quantify the model's accuracy.

The calculated metrics were:

- **Mean Absolute Error (MAE): 0.0005.** Relative to the mean of the target variable 'D' (8.101140), this represents an average absolute error of approximately **0.0062%** (calculated as $(0.0005 / 8.101140) * 100$).
- **Mean Squared Error (MSE): 0.0000.**
- **Root Mean Squared Error (RMSE): 0.0010.** Relative to the mean of 'D' (8.101140), this represents an average squared error magnitude of approximately **0.0123%** (calculated as $(0.0010 / 8.101140) * 100$).
- **R2 Score (Coefficient of Determination): 0.9978.**
- **Explained Variance Score: 0.9979.**

The **efficiency** of the model in explaining the variance in the target variable 'D' is directly indicated by the R2 score and Explained Variance Score. An R2 score of **0.9978** signifies that the model explains approximately **99.78%** of the variance in the target variable 'D' on the test set. Similarly, the Explained Variance Score of **0.9979** suggests that approximately **99.79%** of the variance is explained by the model. These high scores, close to 1, indicate that the trained Gradient Boosting Regressor is highly effective and efficient at capturing the underlying patterns in the data to predict the 'D' variable.

4.5.9 Model Saving and Loading

To enable future use without retraining, the trained `gb_model` was saved to a file named '**gradient_boosting_model.pkl**' using the `joblib` library. `joblib` is suitable for serializing scikit-learn objects. A confirmation message "Model saved successfully!" was printed. The `joblib` version used was **1.5.0**. The model can be loaded back into memory when needed for inference.

4.5.10 Model Prediction (Inference)

The saved model was loaded using `joblib.load()` into the variable `loaded_model`. An example set of input values for the features ('Ws', 'wp', 'wi', 'li', 'VSWR', 'G') was created. These values were formatted into a NumPy array as expected by the model. The `loaded_model` then processed this input array to make a prediction for 'D'. For the provided example input {'Ws': 35278., 'wp': 4.0, 'wi': 1.45, 'li': 0.84, 'VSWR': 1.0275,

'G': 7.599}, the prediction for 'D' was **8.050009342082662**. A user warning about missing feature names during prediction was noted but did not halt the process.

4.5.11 Summary of Machine learning

The mentioned machine learning model training of XGB and other steps are for developing and evaluating a predictive regression model. They describe the initial loading and assessment of a dataset from a CSV file, which have 320 entries, 10 columns, no missing values, and no duplicates. Data cleaning was performed by removing outliers using the IQR method, resulting in a cleaned dataset of 277 rows. Specific features ('Ws', 'wp', 'wi', 'li', 'VSWR', 'G') were selected to predict the target variable 'D'. A Gradient Boosting Regressor model was chosen for the task. The cleaned data was split into 70% for training and 30% for testing. The model was trained on the training data and evaluated on the test data using metrics including MAE, MSE, RMSE, R2, and Explained Variance Score. The evaluation metrics indicated very high performance, with an R2 score of 0.9978 and an Explained Variance Score of 0.9979. Finally, the trained model was saved using joblib for future use and demonstrated with an example prediction.

4.6 Conclusion

Chapter 4 presented a detailed evaluation of the antenna designs culminating in the optimized 4-port MIMO antenna for 5G mmWave applications. Alongside the electromagnetic simulation results showcasing improved bandwidth, gain, isolation, and diversity metrics, a major highlight of this chapter was the development and application of a machine learning model specifically aimed at predicting antenna directivity. The Extreme Gradient Boosting (XGB) regression model was trained on a comprehensive dataset generated from 320 CST simulation iterations, each varying key antenna design parameters. This model achieved high predictive accuracy, with an R^2 score exceeding 99% and a low mean absolute error, demonstrating its capability to reliably estimate directivity without the need for extensive simulation runs. The success of the XGB model not only validates the feasibility of integrating machine learning into antenna design workflows but also offers a powerful tool to accelerate and optimize antenna performance prediction. Building upon these promising results, Chapter 5 will summarize the contributions of the proposed MIMO

antenna design and the effectiveness of the XGB-based directivity prediction model. It will also discuss limitations such as the model's dependency on the quality and scope of training data and the need for further experimental validation. Future work will focus on expanding machine learning applications to other antenna parameters, exploring more advanced algorithms, and integrating real-world measurement data to enhance model robustness. These efforts aim to fully harness the potential of machine learning for rapid, accurate, and cost-effective antenna design in future wireless communication systems.

Chapter 5

Conclusions and Future Work

This study designs, simulates, and uses machine learning to predict the performance of a compact 4-port MIMO antenna optimized for 5G millimeter-wave (mmWave) applications, specifically in the 28 GHz band. The proposed antenna, which is constructed on a Rogers RT/duroid 5880 substrate, is compact (35.58 mm by 35.58 mm) and offers a wide impedance bandwidth of 2 GHz with a center frequency of 27.5 GHz. Performance metrics such as over 29 dB port isolation, 8.32 dB gain, 9.13 dBi directivity, and 98.4% efficiency show that the antenna is suitable for high-speed, low-latency wireless environments. These results validate that the design satisfies the rigorous requirements of next-generation communication systems while being effective in terms of bandwidth, radiation performance, and MIMO compatibility.

Full-wave electromagnetic simulation in CST Microwave Studio was used to develop the antenna, which was refined iteratively. The design changed from a single-element patch antenna to a full 4-port MIMO setup. To fully understand the effects of different design elements like patch length, substrate width, and feedline adjustments, several parametric analyses had been carried out. These adjustments made it possible to optimize impedance matching, resonance behavior, and return loss. The antenna continuously showed wideband performance, strong gain, and steady directional radiation patterns appropriate for mmWave applications throughout each iteration.

Critical metrics were used to carefully evaluate the MIMO performance. Superior diversity performance and little inter-element coupling were shown by the envelope correlation coefficient (ECC), which was less than 0.0007. A channel capacity loss (CCL) of less than 0.1 bps/Hz, a diversity gain (DG) of roughly 10 dB, and a total active reflection coefficient (TARC) below -20 dB were all obtained. These findings support the design's ability to keep high signal efficiency and integrity when multiple ports are excited simultaneously. The antenna's MIMO effectiveness was further increased by maintaining a return loss at the resonant frequency below -30 dB, which ensured low power reflection and superior impedance matching.

A gradient boosting regression model (XGBoost), which integrates machine learning into the antenna design process, was trained to predict antenna directivity based on input design parameters. With an R^2 score of over 99.79%, the model was able to produce extremely accurate predictions and do away with the necessity of running full-wave simulations repeatedly during the design exploration stage. With 320 simulation samples, the model provided a useful substitute for laborious simulations, greatly accelerating the process of performance evaluation. Efficient exploration of the design space was made possible by this data-driven approach, which decreased computational demands without affecting accuracy.

The significant decrease in simulation time and design cost is one of the main benefits of applying machine learning in this situation. Once trained, the machine learning model produces predictions almost instantly, but traditional methods frequently require hours of computation per simulation. Larger design space exploration during optimization stages is made possible by this acceleration. Additionally, the ML models showed that they could capture intricate, nonlinear relationships between geometrical features and performance metrics, which may be difficult for traditional analytical methods to accomplish. The trained model was not only a predictive tool but also facilitated the quick integration of feedback into design iterations, allowing for on-demand or real-time performance assessments.

Despite the successes, the study has some limitations. The antenna was made especially for a limited frequency range (about 28 GHz) and isn't suitable for broadband or multiband applications. Although the compactness is suitable for a wide range of applications, it may still pose integration difficulties in extremely limited settings, such as embedded IoT systems or mobile phones. Furthermore, despite the machine learning model's excellent predictive abilities, it was based on a small number of features and excluded physical phenomena like manufacturing tolerances and surface current behavior.

Furthermore, there are still issues with the ML model's practical deployment. While Google Colab was used to validate the model, neither embedded systems nor commercial CAD tools were used to integrate it. Additional implementation steps would be necessary for real-time usage, such as in interactive simulators or adaptive design platforms. Additionally, data generation was a resource-intensive process; depending on hardware, each CST simulation took about one to two hours. This

limited the size of the training dataset and made dataset compilation laborious. These elements emphasize the need for more accessible and scalable solutions in order to make intelligent antenna design feasible for broad, real-world application.

A number of encouraging possibilities for further research have been identified. Using deep learning techniques like recurrent neural networks (RNNs), convolutional neural networks (CNNs), or artificial neural networks (ANNs) could increase scalability and capture higher-dimensional relationships and more complicated spatial patterns. Inverse design, in which the geometry is predicted based on the desired antenna performance, would also be made possible by this. To increase the model's robustness and generalizability, the dataset should be expanded to include various substrates, slot configurations, and fabrication noise factors. Confidence in the ML framework's practical applicability would also be increased by experimental validation on several physical prototypes in a larger design space.

Lastly, new opportunities can be opened by extending the current design methodology and ML prediction framework to other frequency bands, including 24 GHz, 37–40 GHz, 60 GHz, or even sub-6 GHz bands utilized in early 5G and LTE systems. The application scope would be expanded by investigating various antenna types, including dual-polarized designs, reconfigurable or phased arrays, slot antennas, and monopoles. These additions would put the framework in a position to handle issues in 5G as well as in upcoming fields like IoT networks, V2X (vehicle-to-everything), and 6G communications. In summary, this study establishes a strong basis for the creative development of next-generation wireless antenna systems through a powerful combination of data-driven modeling and electromagnetic simulation.

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